



Glen Dimplex Thermal Solutions

GEHC Platform Integrated Cooling Cabinet Customer Manual





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Important Safety Instructions

1 Important Safety Instructions

This manual contains important safety instructions that should be followed during the installation and maintenance of the **Dimplex Thermal Solutions Integrated Cooling Cabinet (ICC)**. This device has been designed and manufactured specifically for **General Electric Healthcare Company (GEHC)** for integration with GEHC **Magnetic Resonance (MR)** systems. The ICC shall only be used with MR systems that have been approved by GEHC.

Read this manual thoroughly before attempting to install or operate this unit. Failure to follow the instructions in this document may damage the equipment, cause hazardous conditions, and void the warranty. Only properly trained and qualified personnel should move, install, operate, or service this equipment. Adhere to all warnings, cautions and safety instructions on the unit and in this manual when installing, operating, or maintaining the unit. Follow all operating and user instructions.



WARNING

This unit may present arc flash and electric shock hazards that could cause injury or death.

Open all local electric power disconnect switches and wear protective equipment before working within the ICC cabinet.

Earth ground to unit must be provided, per NEC, CEC, and local codes, as applicable. Adhere to all other local codes as applicable.



CAUTION

The pump, tank, and other internal plumbing components can become extremely hot during operation. Allow sufficient time for them to cool before working within the unit. Wear protective gloves and arm protection when working on or near hot components.



NOTICE

Improper installation, application and service practices can result in water leakage from the unit, causing property damage.

Do not locate unit directly above any equipment that could sustain water damage.



NOTICE

Improper storage can cause damage to the unit.

Keep the unit upright, indoors and protected from dampness, freezing temperatures and contact damage.

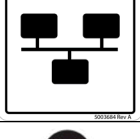


Important Safety Instructions

1.1 Symbol Definitions

Table 1 defines the functional and safety labels used on the ICC. Review the labels thoroughly prior to operating the ICC.

Table 1: Symbol Definitions

Symbol	Description
	Warning: Electrical Hazard
	Warning: Blower Starts Automatically
	Warning: Heavy Weight (Do Not Manually Lift)
	Protective Earth (Ground)
	Read Instruction Manual
	Lock Out/Tag Out before Service
	Ethernet Connection (MODBUS TCP/IP)
	Mass (On Data Tag)



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Table 2: Crating Symbol Definitions

Symbol	Description
	Store between 5% to 95% relative humidity
	Store between -40°C to 70°C ambient temperature
	Center of gravity location

1.2 Regulatory Compliance

The design, construction, and development of the ICC complies with the following national and international regulations:

- Machinery Directive 2006/42/EC
- Low Voltage Directive 2014/35/EU
- Pressure Equipment Directive 2014/68/EU
- IEC 60601-1:2005 +A1:2012 (Ed. 3.1): Medical electrical equipment - Part 1: General requirements for basic safety and essential performance
- IEC 60601-1-2:2014 (Ed. 4): Medical electrical equipment - Part 1-2: General requirements for basic safety and essential performance - Collateral Standard: Electromagnetic disturbances - Requirements and tests
- EN ISO 14971:2019: Medical devices – Application of risk management to medical devices
- General Electric Healthcare Restriction of Hazardous Substances (GEHC RoHS)
- UL 94 Tests for Flammability of Plastic Materials
- UL 796 Printed Wiring Boards
- 29CFR Part 1910.95: FDA Occupational Safety and Health Standards - Audible Noise Regulations
- WEEE directive 2012/19/EU
- 2006/66/EC: European Battery Directive

1.3 Navigating this Document

- Internal document hyperlinks are used throughout for figures, tables headings, etc. (e.g., “refer to Section X.X.X, Figure XX, or Table XX), use **Ctrl + Left Click** to follow that link.
- To navigate back from a hyperlink, use **Alt+ Left Arrow**.



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1.4 Physical Overview

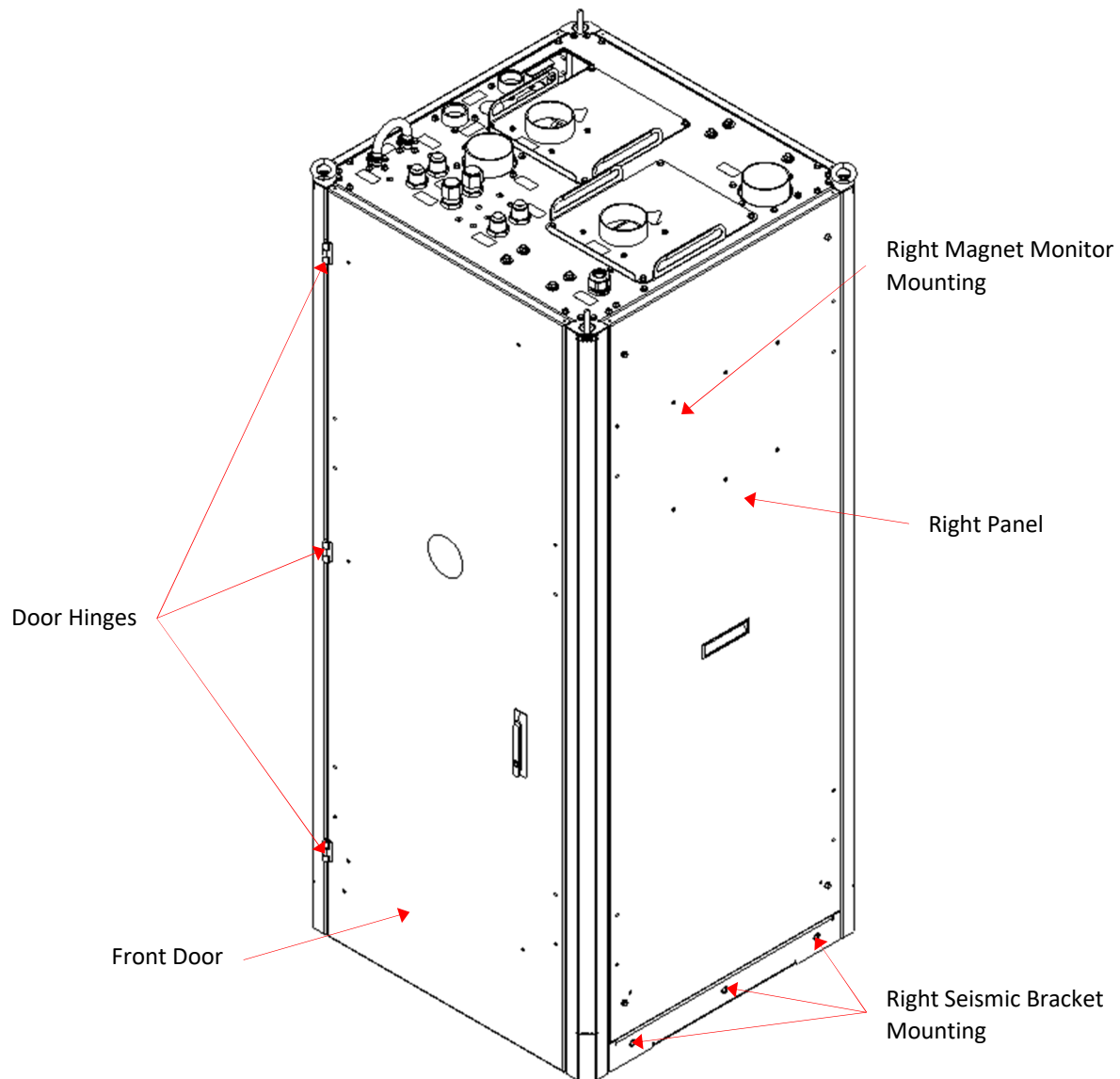


Figure 1-1: Front Isometric View

Important Safety Instructions

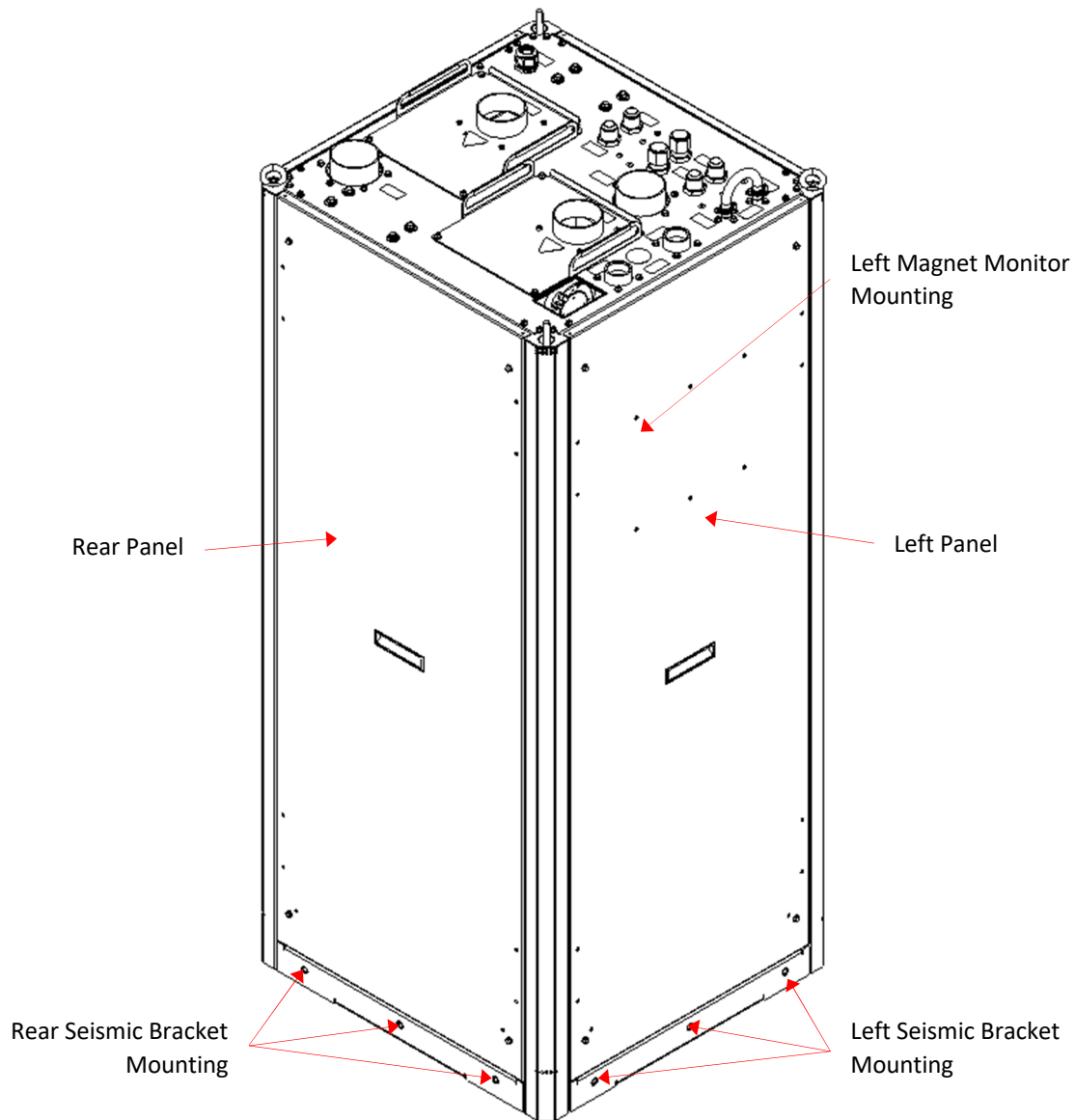


Figure 1-2: Rear Isometric View

1.5 Description

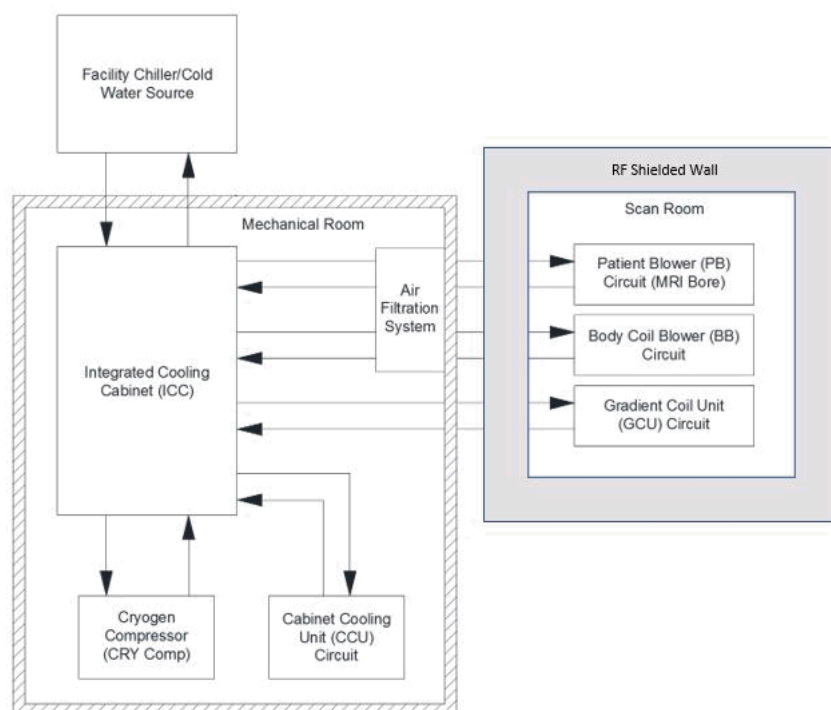


Figure 1-3: ICC as MR Subsystem Block Diagram

The **Dimplex Thermal Solutions Integrated Cooling Cabinet (ICC)** is an automated (controlled by a microprocessor-based, programmable electronic controller) heat exchanger cabinet that facilitates cooling between a process chiller, which is the cooling source, and the magnetic resonance (MR) equipment, which interacts with the patient and produces heat during operation. The ICC receives chilled process fluid from the chiller via the facility plumbing unit (FPU) circuit. This fluid is then diverted between the cold side of the main heat exchanger and the customer provided cryogen compressor. The hot side of the main heat exchanger is connected to two main circuits: the cabinet cooling unit (CCU) circuit and the gradient coil unit (GCU) circuit. The CCU circuit is connected to the MR's integrated systems cabinet (ISC), which consists of the gradient power supplies, amplifiers, and RF amplifiers. The GCU circuit is connected to the MR's gradient coil and a chilled air heat exchanger. Heat is absorbed from these two circuits back to the process chiller via the main heat exchanger.

Temperature control in each circuit is achieved by modulation of individual motorized mixing valves. Flow rate control in each circuit is achieved by modulation of variable frequency drives connected to individual pumps. All control is performed in a closed-loop fashion and managed by a controller.

The ICC also contains two blowers; the body coil blower provides air between the MR's body coil and gradient coil while the patient air blower provides air flow to the MR's patient bore with three selectable



Important Safety Instructions

flow rates. The blowers source air from the scan room and pass through a customer provided filtration system.

1.6 Intended Use



NOTICE

The ICC is fully automatic once fully installed and integrated with the MR system and does not require any manual intervention except for service and periodic maintenance.

The ICC is intended to operate continuously in a stationary mode. It shall be installed in a mechanical room that is separated from the vicinity of the MR and patient. Under the intended use case the ICC is controlled remotely by the customer provided host controller, the Cabinet Monitor, that interacts between the ICC and the MR. The ICC controller sends informational data (temperature, flow rate, pressure, conductivity, etc.) to the host controller and receives commands from it.

The ICC's operational modes correspond to the following GEHC MR system models:

- **Low Flow**
 - SIGNA™ Pioneer 3.0T
 - SIGNA™ Voyager 1.5T
- **High Flow**
 - SIGNA™ Premier
- **Artist-U**
 - SIGNA™ Artist EVO
- **Artist-R**
 - SIGNA™ Artist

Always refer to GEHC's specific documentation for the correct operational mode during installation.

1.6.1 Contraindications

The ICC does not have any patient applied parts and has no known contraindications for its intended use.

1.6.2 Reciprocal Interference

The ICC operates in an isolated mechanical room and has no known reciprocal interferences during the operation of the MR system in its intended use operation.

1.7 Features

- **Individual VFD Driven Centrifugal Pumps**
 - Independent, automatic, closed-loop control and supply of coolant flow rates to process circuits under varying restrictions
 - All stainless-steel wetted components



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- Silicon carbide mechanical seals
- **Motorized Three-way Mixing Valves**
 - Closed-loop control of coolant supply temperatures to process circuits under varying process heat loads
 - 10 second full stroke actuation
 - Separable actuator mechanism for quick replacement
 - Nickel plated body and stainless-steel ball and trim
- **(Air) & Dirt Separator**
 - Purging of entrained air during operation (on CCU and GCU circuits)
 - Particulate filtration of circuits up to 5 microns in size
 - Sediments are removable via integrated ball valve while under use to provide uninterrupted operation
 - Brass construction and stainless-steel mesh
- **CRY Circuit Flow Control Valve**
 - Pressure independent, automatic flow control valve
 - Consistent flow rate supply to CRY circuit under varying FPU coolant supply conditions¹
- **Auxiliary CRY Cooling**
 - City water or auxiliary cooling directly to cryogen compressor in the event of FPU coolant supply loss
- **Deionization Filtration Assembly**
 - Integrated into ICC cabinet for convenient deionization service routines
 - 50-micron particulate filter in series to provide additional filtration
- **Electronic Sensors**
 - All sensors are equipped with connectors at ends for quick field replacements
 - Flow meters
 - Mechatronic (CCU/GCU/FPU circuits) and vortex shedding (CRY circuit) operation
 - Continuous monitoring and reporting of flow rates in all process circuits
 - Used in conjunction with temperature sensors to calculate real-time process heat loads in CCU and GCU circuits
 - Low flow rate protection to prevent pump or process circuit damage
 - Pressure transducers
 - Strain-gage deformation operation
 - Continuous monitoring and reporting of supply pressures in all process circuits
 - High pressure protection to prevent pump or downstream process circuit damage
 - Secondary means of pump flow rate control in the event of flow meter failure

¹ With sufficient FPU supply pressure and flow rate



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- All stainless-steel wetted parts
- Tank level sensor
 - Ultrasonic beam operation
 - Continuous monitoring and reporting of coolant volume in tank
 - Low reservoir warning and fault protection to prevent pumps from running dry or operating with insufficient coolant in system
- Temperature sensor
 - 10k Ohm (at 25°C), Q-curve, NTC thermistor
 - All stainless wetted parts
 - Used in conjunction with flow rate sensors to calculate real-time process heat loads in CCU and GCU circuits
- Conductivity sensor
 - Two electrode conductivity measurement operation with K=0.1 cell constant
 - Integrated 2-wire RTD for temperature compensation
 - Continuous monitoring and reporting of coolant conductivity in CCU and GCU circuits
 - High conductivity warning and error protection to prevent risk of short-circuiting downstream process circuits
- **Centrifugal Blowers**
 - Single speed blower for the body coil circuit
 - Three-speed blower for the patient bore circuit

1.8 Functions

1.8.1 Cooling of Cryogen Compressor

The ICC provides cooling to the customer provided cryogen compressor using coolant provided by the facility chiller in a closed-loop fashion. Alternatively, the cryogen compressor can be cooled using auxiliary coolant (typically city water) in the event of facility chiller failure. To switch to the alternate mode of cooling the isolation ball valves at the alternate cooling supply and return ports are opened and the isolation ball valves of the FPU supply and return branches are closed.

1.8.2 Cooling of the Integrated Systems Cabinet (ISC)

The ICC provides cooling to the MR's Integrated Systems Cabinet (ISC) using coolant in the secondary side of the ICC in a closed-loop fashion. This circuit is referred to as the CCU circuit.

1.8.3 Cooling of the Gradient Coil and Chilled-air Heat Exchanger

The ICC provides cooling to the MR's Gradient Coil and Chilled-air Heat Exchanger using coolant in the secondary side of the ICC in a closed-loop fashion. This circuit is referred to as the GCU circuit.



Installation Guidelines

1.8.4 Integrated Deionization

The ICC provides an integrated deionization planned maintenance internal to the cabinet. The deionization procedure is performed annually to filter impurities out of the cooling system, without needing an external pump or filter chamber.

1.8.5 Air Flow between the Body Coil and Gradient Coil

The ICC provides air flow between the body coil and gradient coil via the ICC's body coil blower. The air is sourced from the MR scan room and filtered by a customer provided filtration system.

1.8.6 Air Flow to the MR Patient Bore

The ICC provides air flow to the MR Patient Bore at three selectable speeds via the ICC's patient blower. The air is sourced from the MR scan room. All air flow to and from the ICC is filtered by a customer provided filtration system.

2 Installation Guidelines

2.1 Mechanical Room

The installation location of the ICC shall be in a mechanical room that is approved by GEHC.

2.2 Personnel, PPE, and Tools

Only properly trained and qualified personnel shall perform the installation, integration, and commissioning of the ICC. All necessary PPE shall always be worn by installation personnel. PPE shall include, but not be limited to:

- Protective helmets
- Safety glasses
- Electrically insulating gloves
- Protective footwear including protection against electrical shock

Only appropriate tools shall be used to install the ICC.

2.3 Installation Procedure

2.3.1 Checking for Shipping Damage

Upon delivery of the ICC always verify that the shipment matches the bill of lading. Inspect the unit immediately for signs of shipping damage both visible and concealed.

2.3.2 Crate Transportation

The ICC is shipped in a crate with protective packing and wrapping that should remain in place until the unit is transported as close to its final location (typically the site's loading area).

When crated, it is recommended that the unit be moved with a forklift or pallet jack from the front or rear of the crate.



Installation Guidelines



WARNING

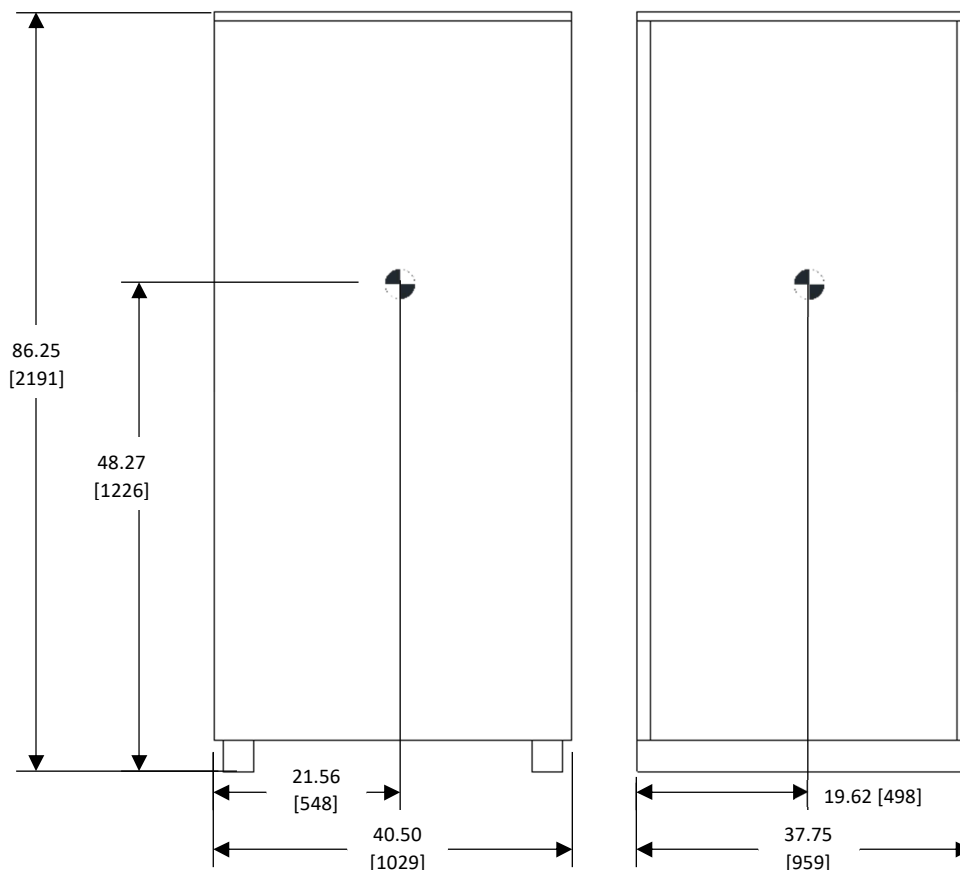
The crate is heavy and there is risk of tilting or falling when moved. Transportation of the crate must only be performed by trained and qualified personnel using appropriate equipment.

Ensure that the ICC is securely positioned (tines of forklift or pallet jack spread as far as permitted along the crate's edge and extend thoroughly through the opposite lifting face) before moving the crate.

Improper handling or insecure lifting of the crate during transportation can cause it to tip and fall leading to injury or death.

If the crate must be rested on an incline, the slope must be no greater than 10°.

The overall dimensions of the crate are shown in Figure 2-1 in inches and millimeters in parentheses. The center of gravity of the crate is also shown and denoted by the  symbol.





Installation Guidelines

Figure 2-1: Crate Overall Dimensions

2.3.3 Uncrating

The disposable packaging is easily uncrated and recycled at the site.



Figure 2-2: pICC Packaging

2.3.3.1 Detaching ICC from Crate Base

The ICC is secured to the base of the crate via two shipping brackets (on the left and right sides).

NOTE: Prior to removing the shipping brackets, make sure the ICC is resting on its feet and not on the wheels, to prevent movement of the ICC when attaching the cabinet dolly.

1. Remove the shipping bracket mounting hardware and crate base fasteners to release the ICC from the base of the crate as shown in Figure 2-6. Once these fasteners have been removed, the ICC can be lifted off the base by either:
2. Using the GEHC MR Cabinet Dolly (GEHC PN 5307574), install three bushings along the bottom of the left and right side of the cabinet.

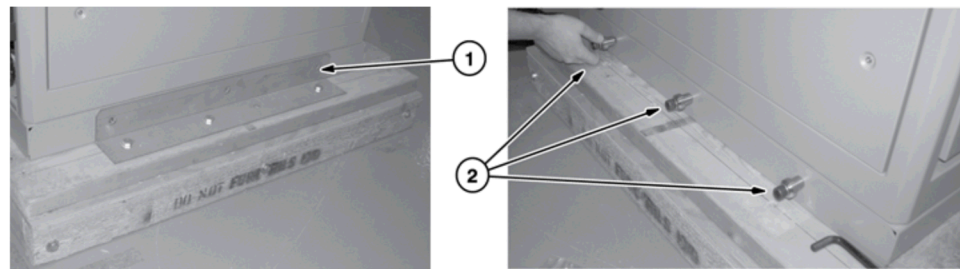


Figure 2-3: Shipping Bracket Removal (Left) and Dolly Bushing Installation Locations (Right)

3. Maneuver the dolly (3) so that all three bushings slip into the slide lock on the attachment plate (4).

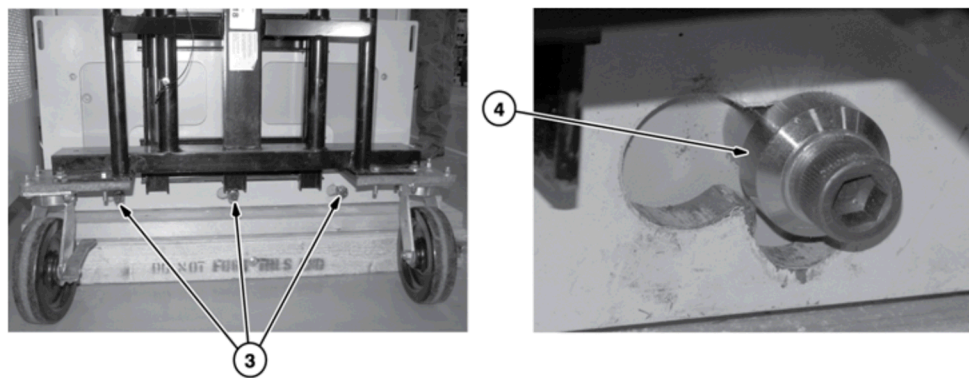


Figure 2-4: Dolly Attachment (Left) and Dolly Jack Screw (Right)

4. Raise the dolly by turning the jack screw clockwise to lock the dolly into position (5).
5. Secure the P-ICC Cabinet to the dollies with the strap (6). Make sure the strap goes through the slot of the attachment plate and the clamp is on a flat surface of the dolly.
6. Tighten the strap and close the clamp (7).
7. Lift the ICC Cabinet off the pallet by turning the jack screw clockwise until the cabinet is raised 1.00 inch (25.40 mm).
8. Alternate raising each dolly in 1/4 in. (6 mm) increments until the cabinet is above the pallet.

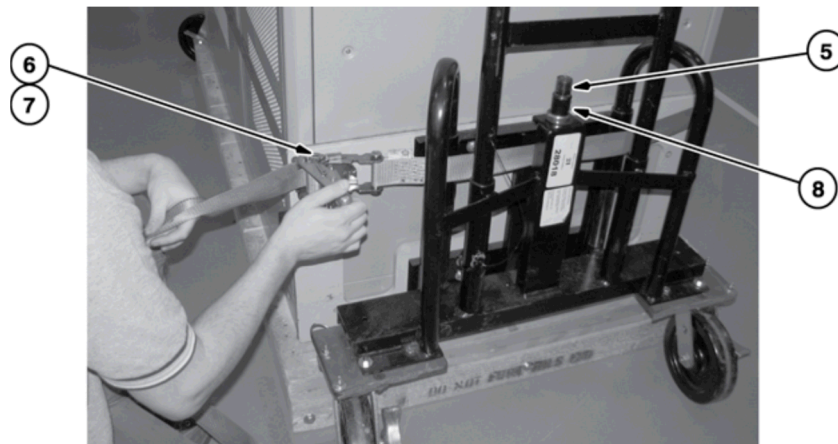


Figure 2-5: Securing the ICC to the Dolly

9. Slide the pallet out from under the cabinet.
10. Alternate lowering each side of the cabinet by turning the jack screw counterclockwise until the cabinet is about 1 in. (25 mm) above the floor.
11. Move the ICC into the Equipment Room.

The ICC can also be lifted vertically via the eyebolts on the lid (refer to Figure 2-7)

- **Ensure that chains or straps used are rated for the weight of the ICC.**
- **Ensure that chains or straps used are securely fastened to the eyebolts and the lifting mechanism**

Installation Guidelines

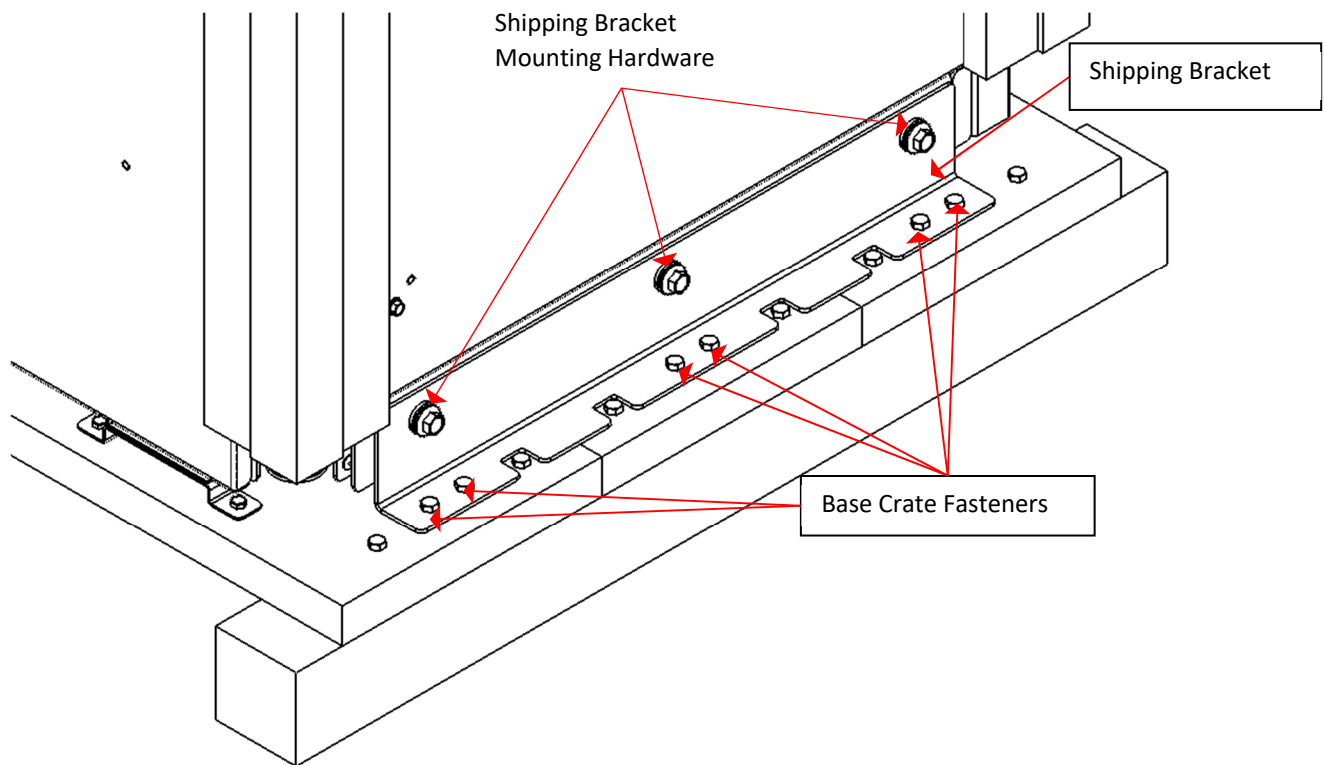
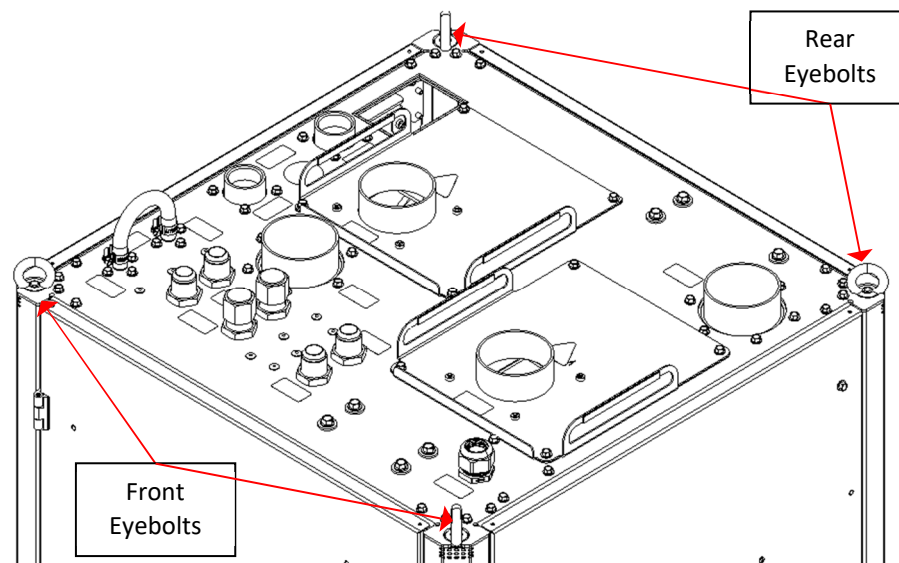
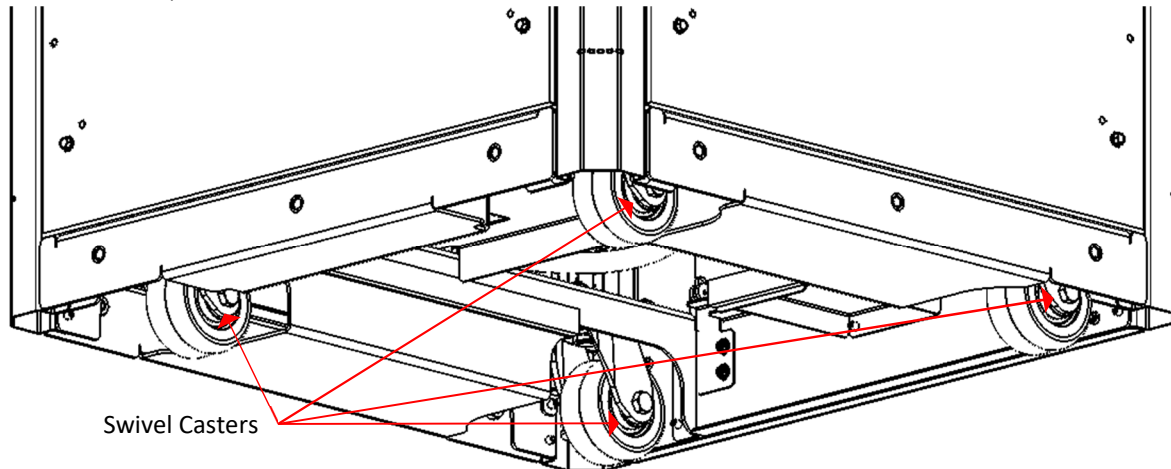


Figure 2-6: Crate Base Mounting



*Figure 2-7: Lifting Eyebolts***2.3.4 ICC Transportation***Figure 2-8: Swivel Casters at Bottom*

The ICC has four swivel casters that can rotate 360° to allow the ICC to be pushed to its final installation location.

**WARNING**

The ICC is top-heavy and there is risk of tilting or falling when moved.

If the ICC must be pushed along an incline on its casters the slope must be no greater than 11°.

**NOTICE**

To ensure proper installation of the cryogen compressor it might be necessary to orient the casters such that they point towards the front or rear of the ICC for the maximum possible clearance. This can be achieved by lifting the casters off the ground using the lifting feet (refer to 2.3.5.2).

2.3.5 Internal Access

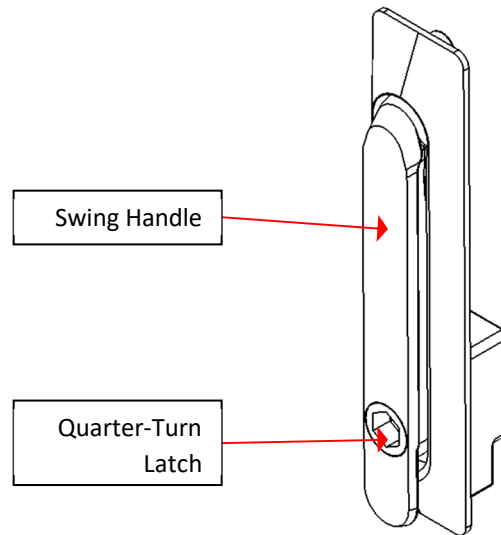


Figure 2-9: Front Door Swing Handle Access

To open the ICC front door and access internal components:

1. Use a 10mm male hex (Allen) wrench to unlatch (turn counterclockwise) the Quarter-Turn Latch to release the Swing Handle.
2. Lift the Swing Handle and turn counterclockwise to retract the door rods.
3. Pull the door outwards (from right to left when facing the ICC's front). The door can be opened to 180° if necessary.

To close the ICC front door and lock:

1. Ensure that the Swing Handle is in its fully open position (refer to Figure 2-9).
2. Push the door to its closed position until it is flush with the cabinet.
3. Turn the swing handle clockwise and push into its recessed position to extend the door rods.
4. Use a 10mm male hex wrench to latch (turn clockwise) the Quarter-Turn Latch to lock the Swing Handle (refer to Figure 2-9).



WARNING

The front door must be fully closed and properly secured when the ICC is operational in the absence of service personnel operating on the ICC.

2.3.5.1 Door Removal

The ICC front door is connected to the cabinet frame by two separable hinges. The hinges can be separated to remove the front door from the cabinet frame. Remove the front door as follows:

1. Access the inside of the left side of the cabinet base (refer to Figure 2-10).
2. Use a 10 mm wrench to uninstall the M6 Ground Strap Nut.
3. Release the Ground Strap Terminal from the Ground Stud.
4. Loosely reinstall the Ground Strap Nut onto the door's ground stud to prevent misplacing it. The Tooth Lock Washer can remain in place.
5. Locate the three Hinges (refer to Figure 1-1) and lift the Hinge Pin of each hinge upward (a rubber mallet may be required).
6. Release the Door from each Hinge as shown in Figure 2-11.

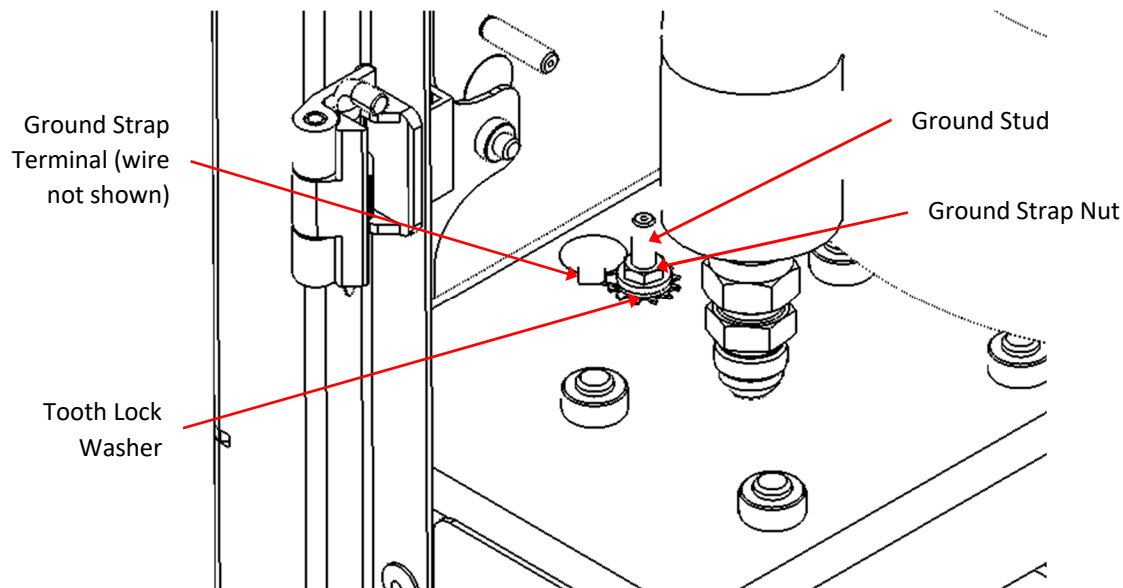


Figure 2-10: Ground Strap

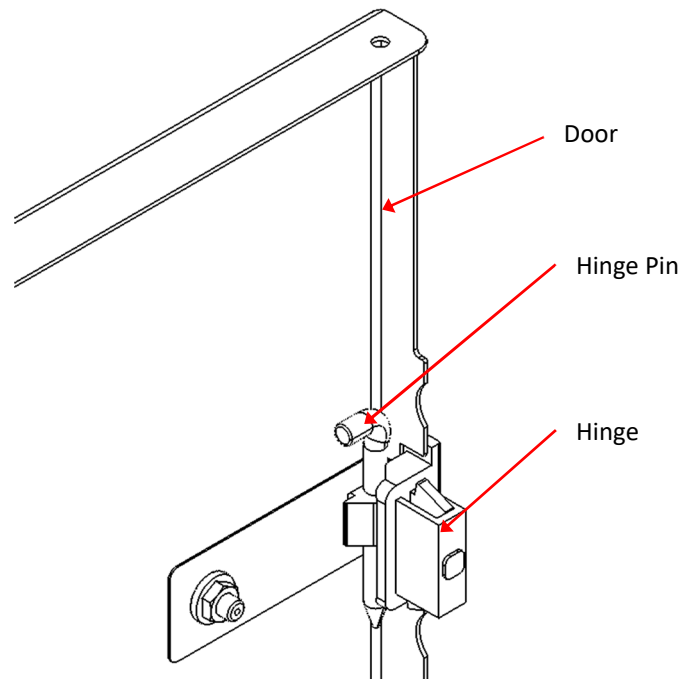


Figure 2-11: Door Hinge and Pin Installation/Removal

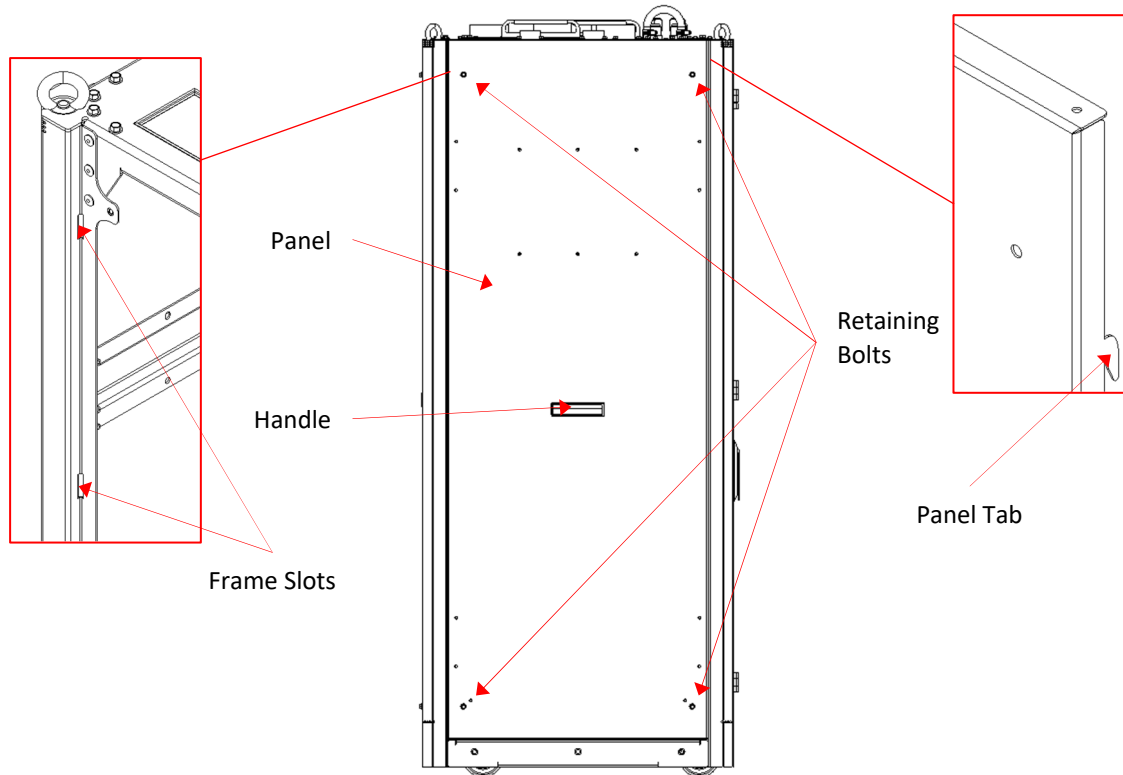
To reinstall the door to the hinges:

1. Align the Door to each Hinge as shown in Figure 2-11.
2. Insert the Hinge Pins into each Hinge's hole securely to fasten the Door to the Hinges (a rubber mallet may be required).
3. Use a 10mm wrench to uninstall the M6 ground strap nut from the Ground Stud (refer to Figure 2-10) if not previously installed.
4. Install the Door's Ground Strap Terminal onto the Ground Stud.
5. Install the Ground Strap Nut over the Ground Terminal.



WARNING

Ensure that the ground strap is properly installed and securely fastened when the door is reinstalled. Failure to do so may result in improper electrical grounding of the door and a risk of electric shock that may lead to injury or death.

2.3.5.2 Panel Removal*Figure 2-12: Panel Removal (Left View)*

The side (left and right) and rear panels are removable if access from other sides of the ICC is required.

To remove a panel

1. Locate the panel to be removed.
2. Use a 10mm wrench to unfasten the four M6x1 Retaining Bolts and retain (refer to Figure 2-12).
3. Lift the panel, by the Handle, upwards and outwards to release the Panel Tabs from the Frame Slots (refer to Figure 2-12). Carefully set the removed panel against the ICC as is still connected to the frame by the Ground Strap.
4. Access the ICC through the front door (refer to section 2.3.5).
5. Locate the Ground Strap connected to the removed panel and uninstall (refer to Figure 2-10):
 - a. Use a 10 mm wrench to uninstall the M6 Ground Strap Nut.
 - b. Release the Ground Strap Terminal from the Ground Stud.
 - c. Loosely reinstall the Ground Strap Nut onto the door's ground stud to prevent misplacing it. The Tooth Lock Washer can remain in place.

To reinstall a panel



Installation Guidelines

1. Lift the panel by the Handle and align all Panel Tabs with corresponding Frame Slots. Push the panel inward until the tabs clear the slots and release. A rubber mallet may be used gently tap the top of the panel to align it correctly.
2. Access the ICC through the front door (refer to section 2.3.5).
3. Locate the Ground Strap previously uninstalled from the removed panel and reinstall (refer to Figure 2-10):
 - a. Install the Panel's Ground Strap Terminal onto the Ground Stud.
 - b. Install the Ground Strap Nut over the Ground Terminal.

2.3.6 Leveling Feet

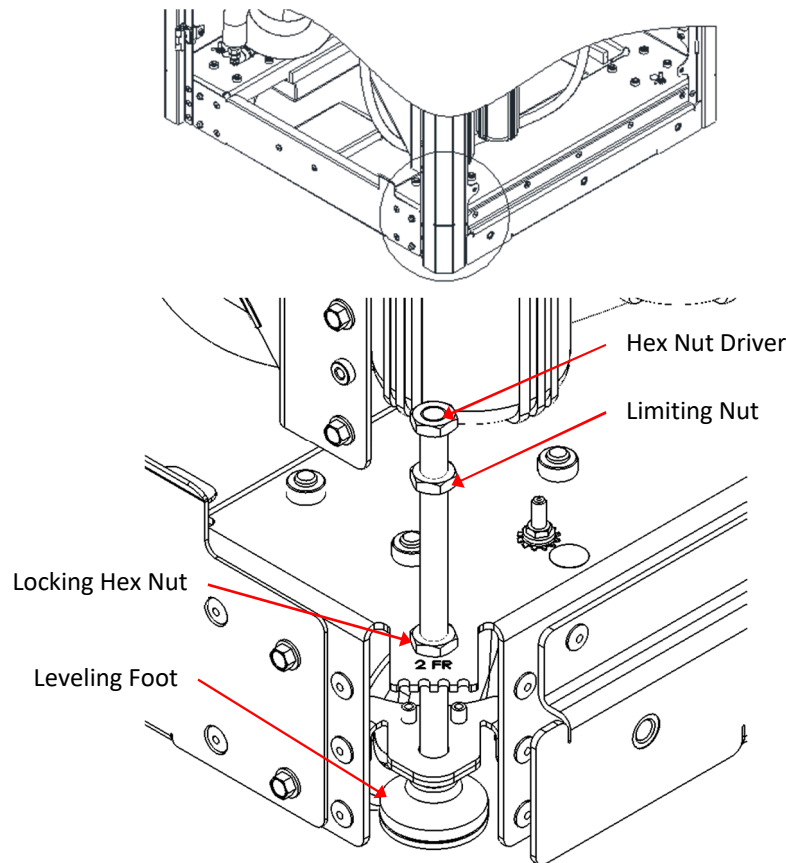


Figure 2-13: Leveling Feet

The ICC has four leveling feet that are used to level the ICC. Lower the four leveling feet on the ICC so that they reach the ground. To install the leveling feet:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Locate the four Threaded Rods in each corner (one shown in Figure 2-13).



Installation Guidelines

3. Use a 19mm wrench to turn the Locking Hex Nut (refer to Figure 2-13) counterclockwise to free the threaded rod. Repeat on all required rods as necessary.
4. Use a 19mm wrench to turn the Hex Nut Driver (refer to Figure 2-13) on each Threaded Rod gradually clockwise in an alternating fashion (no more than two full rotations per rod at once) until the ICC is leveled or lifted to the desired height. The Locking Hex Nut may need to be further recessed up the Threaded Rod. Repeat on all required rods as necessary.



WARNING

The limiting hex nut prevents the feet from extending beyond 30mm off the ground. If the cabinet must be lifted at this maximum height, **ensure that the incline of the floor is less than 15° to prevent tipping and falling over of the ICC.**

2.3.7 Seismic Brackets



NOTICE

For seismic areas the customer's contractor must provide and install brackets that secure the P-ICC cabinet to the floor, in the location shown below, secured to the M12 threads in the cabinet.

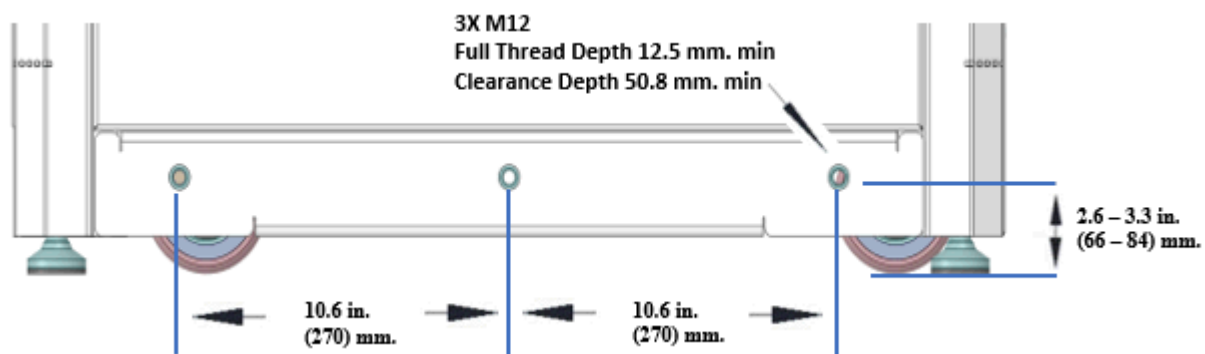


Figure 2-14: Seismic Bracket Mounting Dimensions

The ICC allows for the installation of seismic brackets onto the left, right and rear sides of the ICC. At minimum, seismic brackets must be installed on the left and right side of the ICC. To install the seismic brackets:

1. Align the seismic bracket as shown in Figure 2-14 with the mounting nuts on the desired side of the ICC.
2. Install the seismic bracket to the ICC using the hardware supplied with the shipping bracket
3. Install the seismic bracket to the ground using hardware procured and installed locally per site architectural drawings.

4. Repeat steps 1 to 3 for the other sides as necessary.

2.3.8 Cryogen Compressor

The ICC is capable of housing a Sumitomo F-50 or F-50H cryogen compressor within.

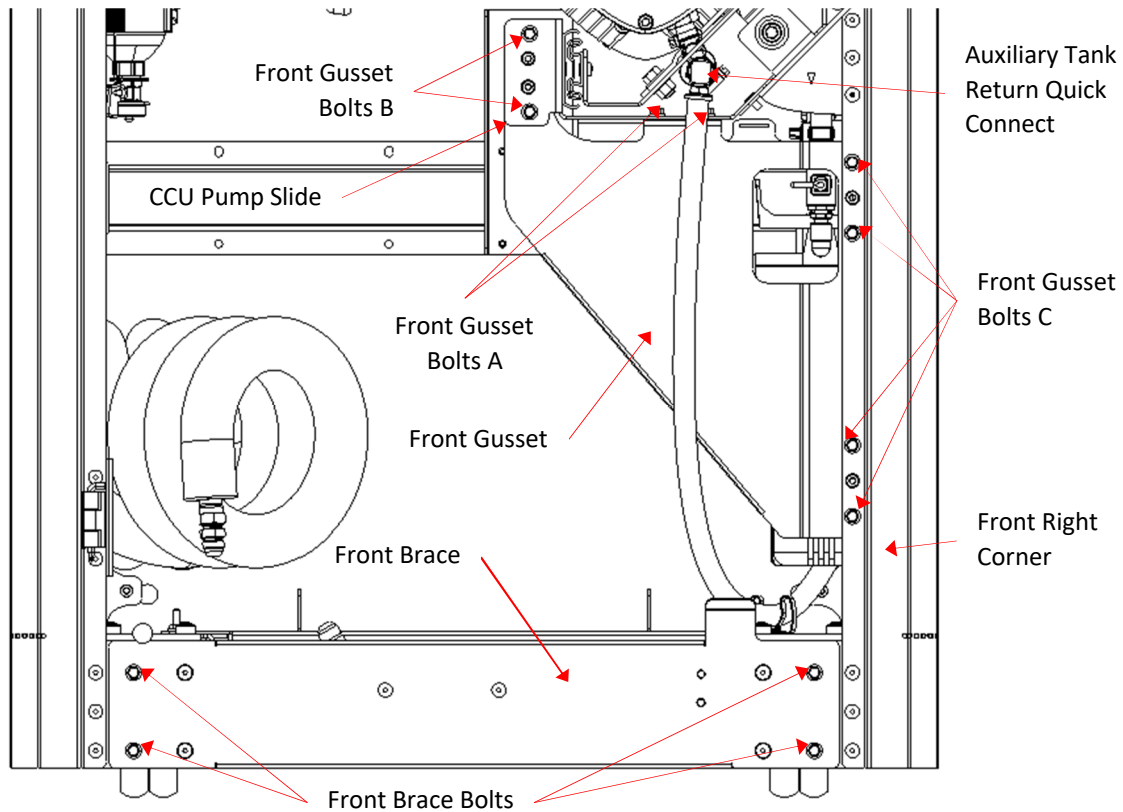


Figure 2-15: Cryogen Compressor Installation (Front View)

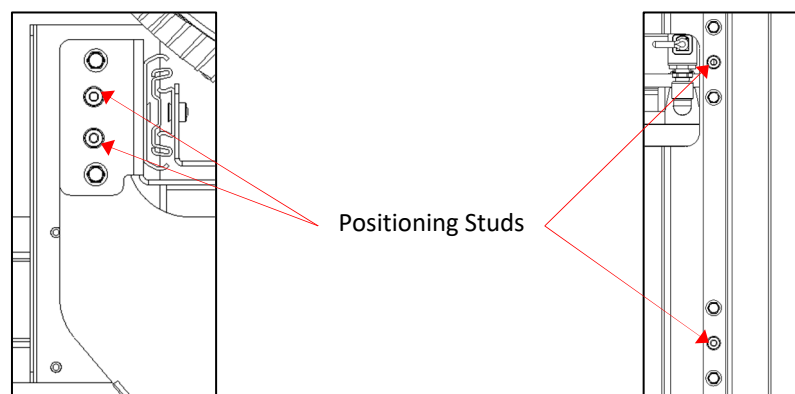


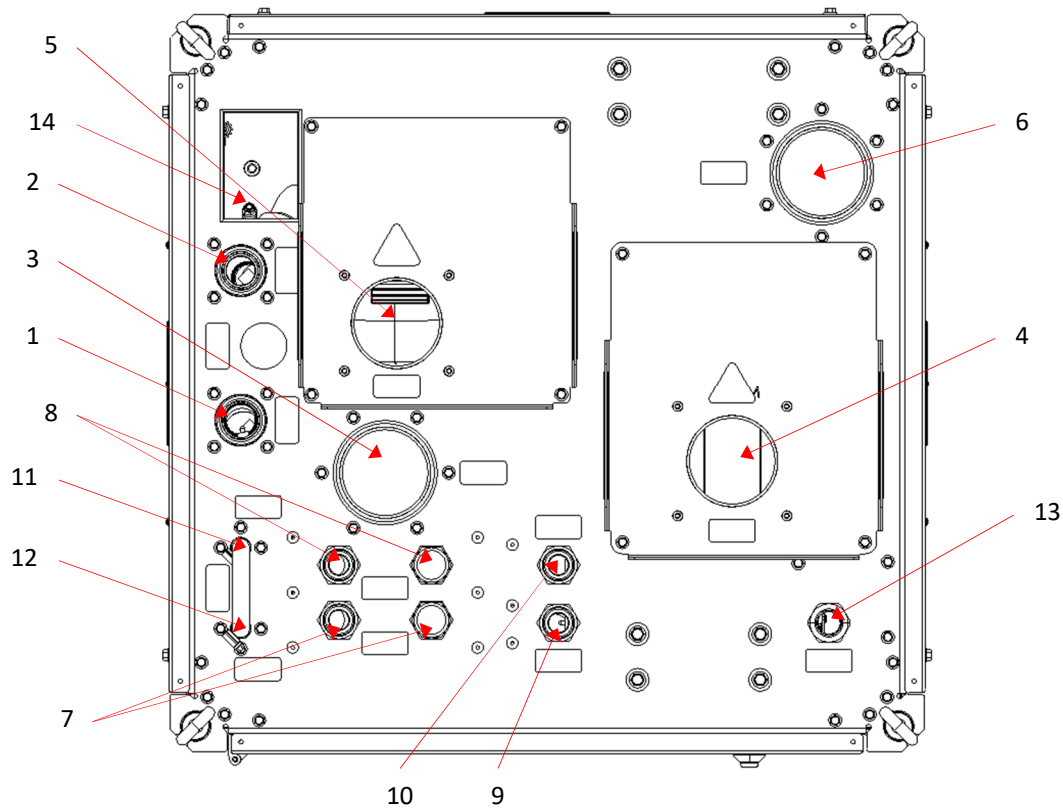
Figure 2-16: Positioning Studs for Front Gusset



Installation Guidelines

To install the cryogen compressor in the ICC:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Disconnect the Auxiliary Tank Return Quick Connect hose.
3. Uninstall the Front Gusset (refer to Figure 2-15):
 - a. Use a 10mm wrench to unfasten the two M6 Front Gusset Bolts A.
 - b. Use a 10mm wrench to unfasten the two M6 Front Gusset Bolts B.
 - c. Use a 10mm wrench to unfasten the four M6 Front Gusset Bolts C.
 - d. Remove the Front Gusset.
4. Uninstall the Front Brace (refer to Figure 2-15)
 - a. Use a 10mm wrench to unfasten the four M6 Front Brace Bolts.
 - b. Remove the Front Brace.
5. Remove the Ship Loose Box and Ship Loose Bracket according to 2.4. Proceed to step 6 if these have previously been removed.
6. Maneuver the cryogen compressor into the ICC opening. The front of the cryogen compressor should face the front of the ICC. The cryogen compressor should contact foam strips in the ICC cabinet at each of its three edges (left, right and rear).
7. Install the cooling water plumbing connections, power, and helium to and from the cryogen compressor according to Install Manual.
8. Reinstall the Front Brace and its bolts. The lower door stop of the brace should face upwards (refer to Figure 2-15).
9. Reinstall the Front Gusset and its bolts to the CCU Pump Slide Base and Front Right Corner (refer to Figure 2-16). **Note:** The Front Gusset fits behind the CCU Pump Slide Base and Front Right Corner.
 - a. Align the Front Gusset with the positioning studs shown in Figure 2-16
 - b. Use a 10mm wrench to install the two M6 Front Gusset Bolts B.
 - c. Use a 10mm wrench to install the four M6 Front Gusset Bolts C.
 - d. Use a 10mm wrench to install the two M6 Front Gusset Bolts A.
10. Reinstall the Auxiliary Tank Return Quick Connect hose to the CCU Pump Drain Port (refer to Figure 2-15).

2.3.9 Connections**2.3.9.1 Lid Layout***Figure 2-17: Lid Connection Diagram**Table 3: Lid Connection Table*

Number	Label	Type	Function
1	FAC PRIM SPY	1-1/2" FPT	FPU coolant supply from chiller to ICC
2	FAC PRIM RTN	1-1/2" FPT	FPU coolant return from ICC to chiller
3	BC AIR RTN	4" Hose Duct	BB air return from MR to ICC
4	BC AIR SPY	4" Hose Duct	BB air supply from ICC to MR
5	PAT AIR SPY	4" Hose Duct	PB air supply from ICC to MR
6	PAT AIR RTN	4" Hose Duct	PB air return from MR to ICC
7	ISC SEC RTN	1" 37° JIC Male (Size 16)	CCU coolant return from MR to ICC
8	ISC SEC SPY	1" 37° JIC Male (Size 16)	CCU coolant supply from ICC to MR
9	GC SEC RTN	1" 37° JIC Male (Size 16)	GCU coolant return from MR to ICC
10	GC SEC SPY	1" 37° JIC Male (Size 16)	GCU coolant supply from ICC to MR
11	CW RTN	1/2" Hose Bead	Emergency backup coolant supply to ICC CRY circuit



Installation Guidelines

12	CW SPY	1/2" Hose Bead	Emergency backup coolant return from ICC CRY circuit
13	J1	Cord Grip	Incoming power cable to ICC
14	N/A	Opening	Helium hose connections to and from cryogen compressor, serial cable (44-pin D-sub) connection to electrical enclosure, PHPS2 signal cable to PB

2.3.9.2 Incoming Power



WARNING

This procedure involves a risk of electric shock that could cause property damage, injury and/or death. All electrical connections should be performed only by properly trained and certified technicians wearing proper protective gear and using properly insulated tools. Before attempting to make any electrical connections or disconnections to the ICC:

- **Verify that the incoming power to the ICC has been turned off**
- **Lock Out and Tag Out main electrical connecting points**
- **Use a voltmeter to verify that there is no incoming power to the ICC**

The ICC shall receive electrical power only from a GEHC approved Power Distribution Unit (PDU) at the installation site.



WARNING

The ICC is a Class I device; to avoid the risk of electric shock, this equipment must only be connected to a supply main with protective earth.

Connect power to the ICC as follows:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Located the Power Distribution Enclosure (refer to Figure 2-18). Use a 3mm hex wrench and unfasten the four M5 Cover Bolts.
3. Route the incoming power cable (GEHC cable run number E0007) from the top through the J1 cord grip (refer to Figure 2-17 and Table 3) into the Power Distribution Enclosure.
 - The maximum diameter of the cable must not exceed 25.4mm
4. Install each individual incoming power cable wire (excluding ground) onto the upper row of the incoming power terminal block (refer to Figure 2-19)
 - The clearance between each terminal post is 14.2 mm
 - The terminal block screws are #10-32 with a Phillips drive
 - Tighten until snug
5. Use a 10mm wrench to unfasten the top M6 Ground Nut (refer to Figure 2-19). The bottom M6 Ground Nut does not need to be uninstalled. Install the ground wire terminal over this stud. Reinstall the top Ground Nut.

6. Ensure that upstream power (at the Power Distribution Unit) has been turned off. Ensure that the Main Power Circuit Breaker is turned on (indicator will be red, refer to Figure 2-19).
7. Reinstall the Cover and Cover Bolts.

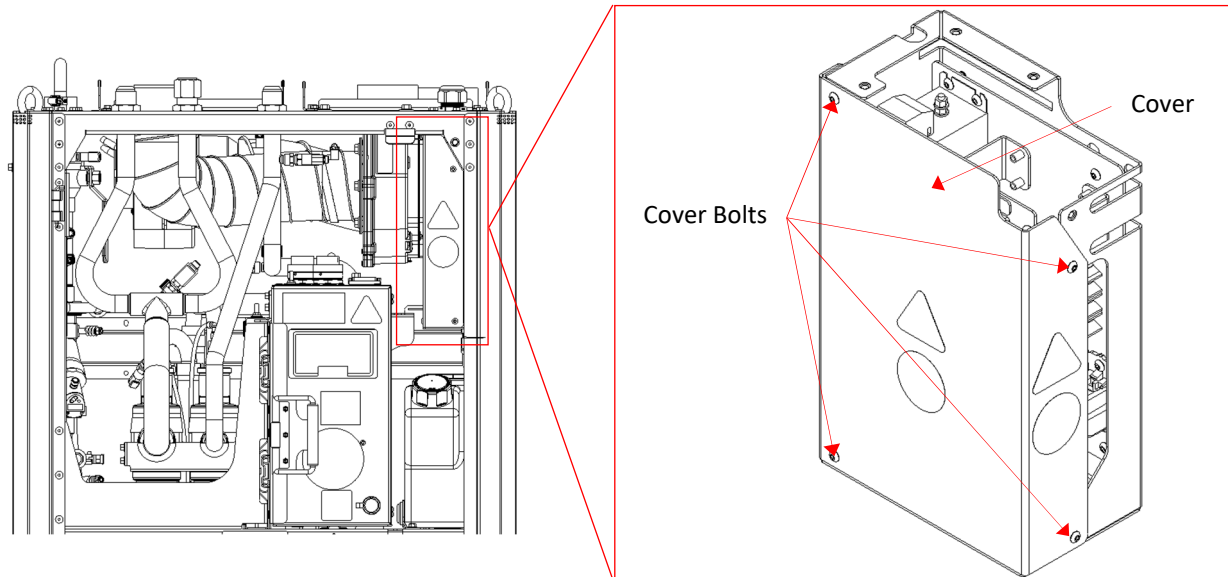


Figure 2-18: Power Distribution Enclosure Location and Fasteners

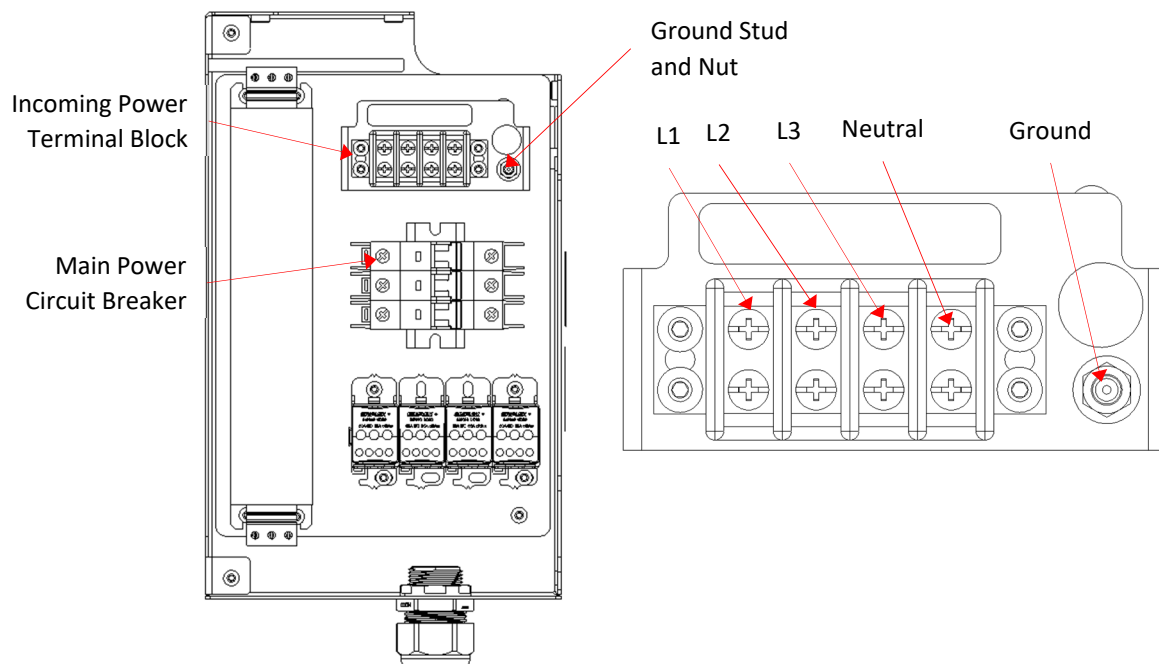


Figure 2-19: Incoming Power into Terminal Block



Installation Guidelines



WARNING

All internal component wiring has been verified to be correct at the factory. **Do NOT swap any of the ICC's components as this will lead to incorrect motor rotation, damage the ICC, and void warranty.**

2.3.9.3 Cabinet Monitor Signal (44-pin)

The ICC passes bi-directional signals with the MR Cabinet Monitor through a 44-pin D-sub serial cable (GEHC cable run number E3040).



NOTICE

The ICC can be operated (for initial priming of the secondary plumbing circuits) without this connection by using the provided 44-pin D-sub jumper (DTS PN 710205). If this is desired, this process can be skipped until this 44-pin D-sub serial cable is ready for installation (refer to 2.3.12).

Connect this cable to the ICC as follows:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Route the 44-pin serial cable through the opening in the rear of the lid, 14 (refer to Figure 2-17) towards the Control Enclosure (refer to Figure 2-20).
3. Connect the cable to the J2 connector (refer to Figure 2-20). Note: This connector is keyed so only one orientation is possible.
4. Fasten the jack screws of the cable to the Jack Screw Nuts on the J2 connector (refer to Figure 2-21).

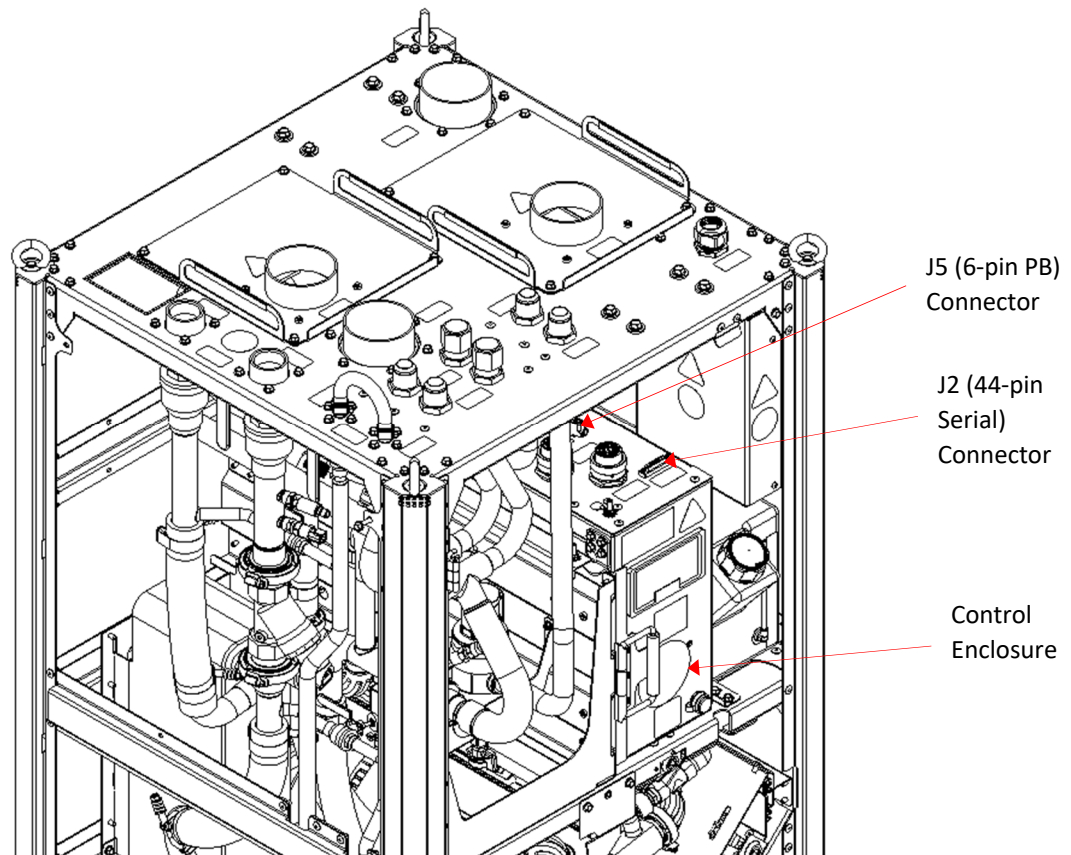


Figure 2-20: Control Enclosure and J2 and J5 Connectors

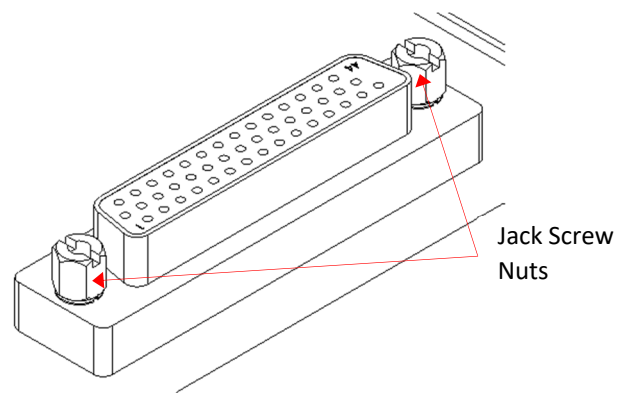


Figure 2-21: J2 Jack Screw Nuts

2.3.9.4 Patient Blower Signal (6-pin)

The PHPS2 controls the speed of the Patient Blower via a 6-pin cable (GEHC cable run number E0010). Connect this cable to the ICC as follows:



Installation Guidelines

1. Access the ICC through the front door (refer to section 2.3.5).
2. Route the 6-pin cable through the opening in the rear of the lid, 14 (refer to Figure 2-17) towards the Control Enclosure (refer to Figure 2-20).
3. Connect the cable to the J5 connector (refer to Figure 2-20). Note: This connector is keyed so only one orientation is possible.

2.3.9.5 Plumbing Connections



WARNING

This procedure involves a risk of coolant leakage that could cause property damage, injury and/or death. All plumbing connections should be performed only by properly trained and certified technicians wearing proper protective gear and using proper tools.

Always ensure that external plumbing connections have been closed/turned off prior to installing or uninstalling plumbing connections.

Perform ALL plumbing connection installation procedures in this section (2.3.9.3) prior to opening/turning on external plumbing connections.

Always ensure that plumbing connections are free of leaks after installation to prevent property damage or hazardous slip and fall conditions that may lead to injury of death.

2.3.9.5.1 Cryogen Compressor Cooling

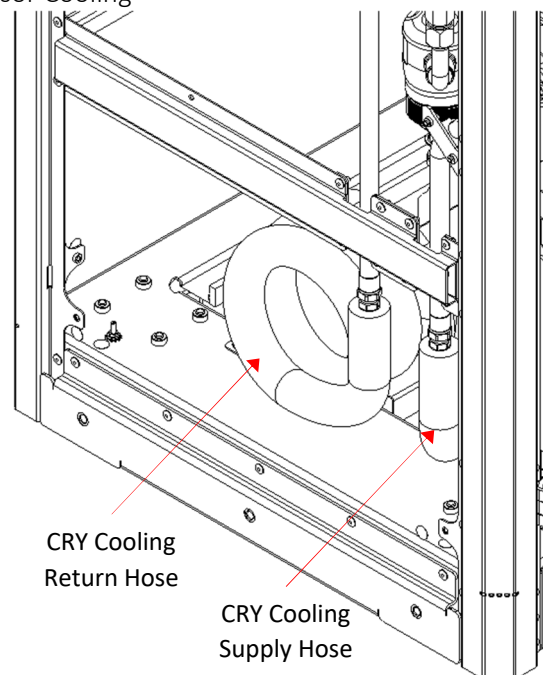


Figure 2-22: Cryogen Compressor Cooling Hoses

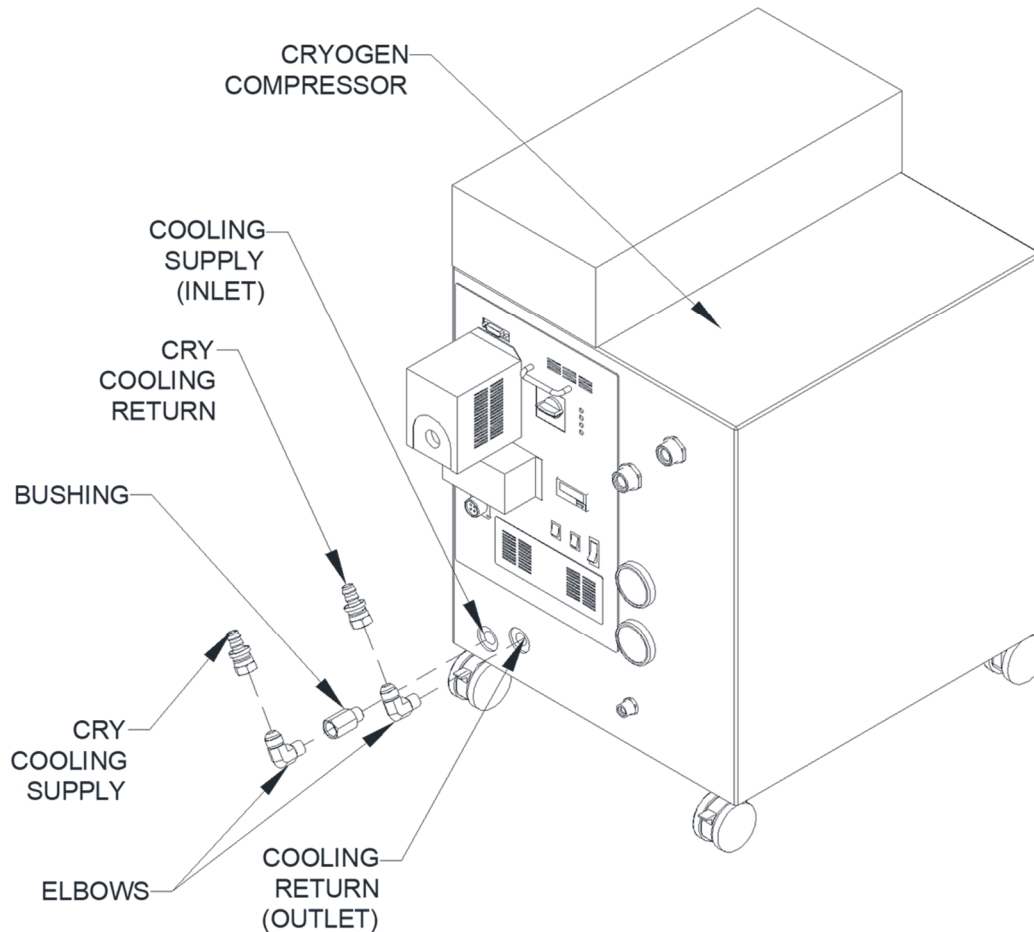


Figure 2-23: Cryogen Compressor Cooling Plumbing Connections

The cryogen compressor fittings kit is supplied within the ship loose box (refer to 2.4. The kit contains the following:

- **Elbows:** Two (2) stainless steel 90° 1/2" (size -8) JIC male adapter to 3/8" BSPT male elbow adapters
- **Bushing:** One (1) stainless steel 3/8" BSPT male to female bushing

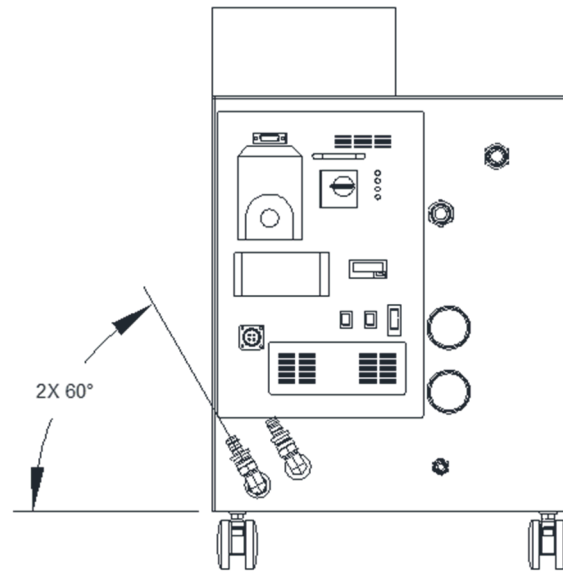


Figure 2-24: Cryogen Compressor Cooling Plumbing Orientation

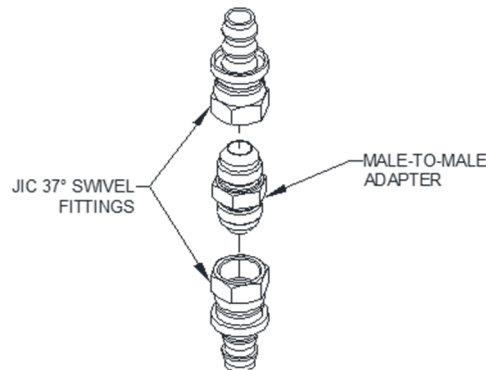


Figure 2-25: Cryogen Compressor Cooling Jumper Connection Removal

To install the cryogen compressor cooling connections:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Locate the CRY Cooling Hoses (refer to Figure 2-22). Clip the cable ties on each hose to unravel them. The hoses are connected from the factory by a male-to-male adapter (jumper connection) (refer to Figure 2-25). Use a 7/8" wrench (for the JIC 37° hose swivel connectors) and a 13/16" wrench (for the male-to-male adapter) and disconnect the hoses from each other and the adapter. The male-to-male adapter can be discarded.
3. Use a 19mm wrench to install an Elbow at the Cooling Return port on the cryogen compressor. Orient this fitting as shown in Figure 2-24. Install the Cry Cooling Return Hose (labeled **CRY PRIM RTN**) to the other end of this elbow.

4. Use a 7/8" wrench to install the Bushing at the Cooling Supply port on the cryogen compressor. Install an Elbow onto the other end of the Bushing. Orient this fitting as shown in Figure 2-24. Install the Cry Cooling Supply Hose (labeled **CRY PRIM SPY**) to the other end of this elbow.

**NOTICE**

Always consult the cryogen compressor's installation manual for the correct installation procedures. Always verify that the cooling water supply and return labels are correct prior to installation.

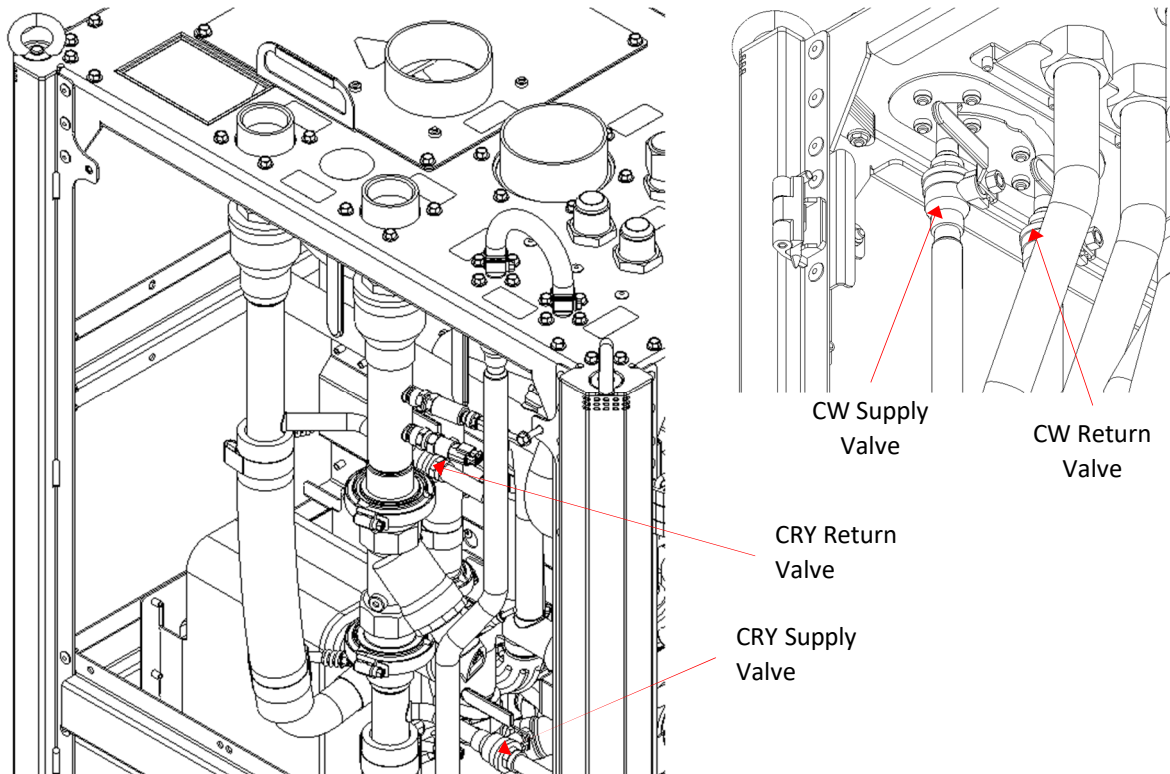


Figure 2-26: Cryogen Compressor Cooling Valves for Adjustment

Configure the valves in the ICC as follows to enable coolant flow to the cryogen compressor:

1. Locate the CRY Supply and Return Valves (refer to Figure 2-26). Open both valves (handle parallel to direction of flow).
2. Locate the CW Supply and Return Valves (refer to Figure 2-26). Ensure both valves are closed (handle perpendicular to direction of flow).

2.3.9.5.2 Facility Plumbing Connections

**WARNING**

The supply pressure for the facility side circuit, measured at the inlet of the ICC, must **NOT** exceed **6 bar**.

- Each FPU port is a 1-1/2" female NPT.
- Connect the facility supply (typically chiller) to the ICC port labeled **FAC PRIM SPY** on the lid (refer to Figure 2-17)
- Connect the facility return (typically chiller) to the ICC port labeled **FAC PRIM RTN** on the lid (refer to Figure 2-17)
- Wrap PTFE (Teflon) tape, one-and-a-half wraps around the male portion of the thread. Start two threads up from the end of the male portion of the fitting.
- Install NPT plumbing connections per **ASME B1.20.3** standards (refer to 8.1).

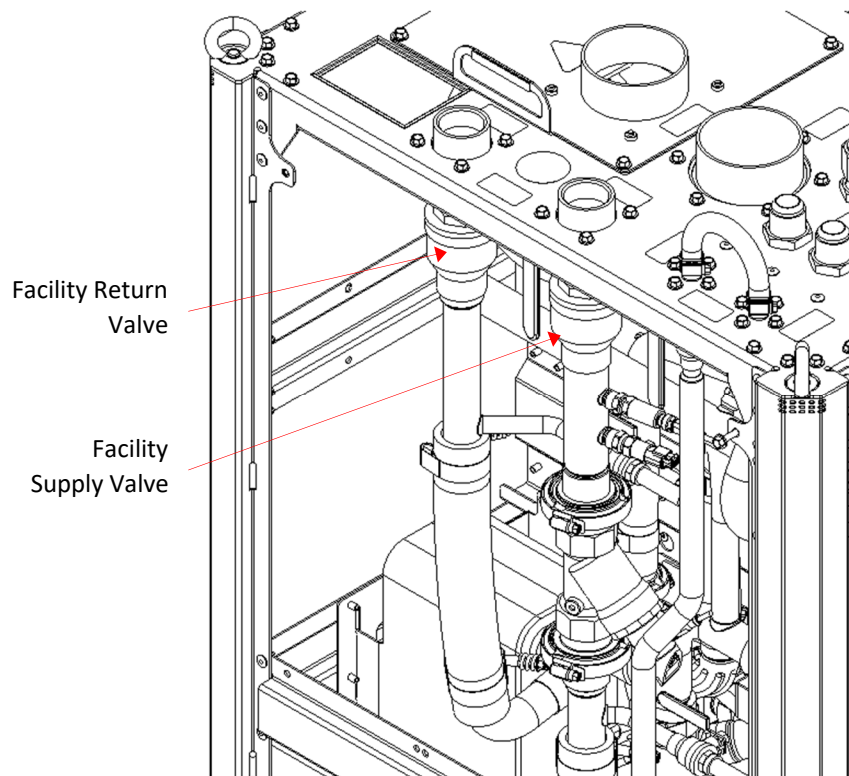


Figure 2-27: FPU Valves for Adjustment

Configure the valves in the ICC as follows to enable coolant flow to the FPU circuit:

1. **Ensure that all steps in section 2.3.9.5.1 have been properly completed.**
2. Locate the FPU supply and return valves in the ICC (refer to Figure 2-27).



Installation Guidelines

- a. Open both valves (handle parallel to direction of flow).

2.3.9.5.3 CCU Plumbing Connections

- Connect the MR ISC supply to the ICC port labeled **ISC SEC SPY** on the lid (refer to Figure 2-17).
- Connect the MR ISC return to the ICC port labeled **ISC SEC RTN** on the lid (refer to Figure 2-17).
- Install JIC 37° flare connections until light wrench-tight then 1/6th of a turn.



NOTICE

- For **low flow applications** use one of each CCU supply and return port. Leave factory installed JIC nut cap in place.
- For **high flow applications** use both CCU supply and return ports. Remove the factory installed JIC nut cap.

2.3.9.5.4 GCU Plumbing Connections

- Connect the MR GCU supply to the ICC port labeled **GC SEC SPY** on the lid (refer to Figure 2-17).
- Connect the MR GCU return to the ICC port labeled **GC SEC RTN** on the lid (refer to Figure 2-17).
- Install JIC 37° flare connections until light wrench-tight then 1/6th of a turn.

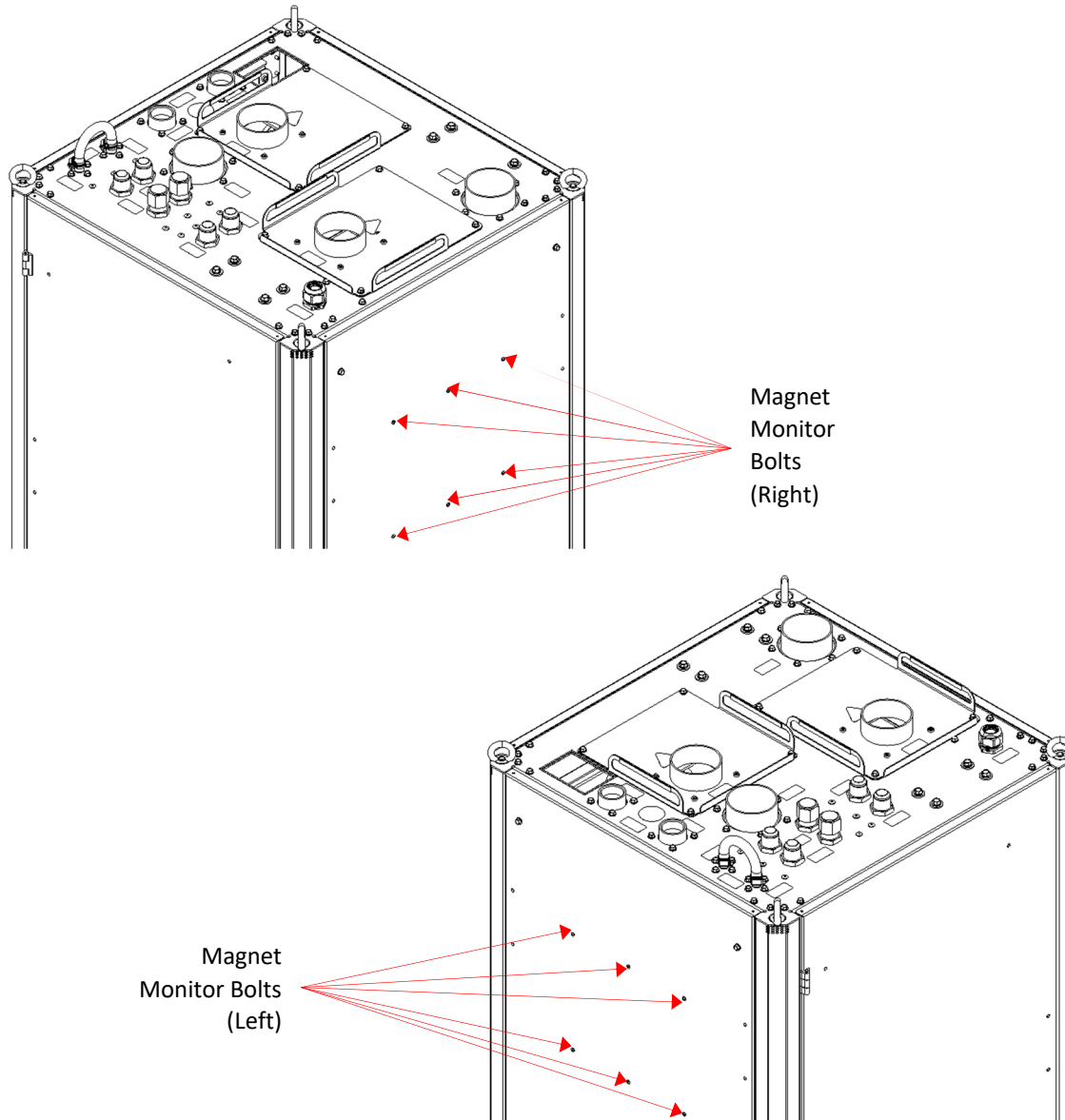
2.3.9.6 Blower Connections

2.3.9.6.1 Body Coil Blower

- Connect the MR body coil blower supply to the ICC port labeled **BC AIR SPY** on the lid (refer to Figure 2-17).
- Connect the MR body coil blower supply to the ICC port labeled **BC AIR RTN** on the lid (refer to Figure 2-17).
- Install air hose clamp connections until snug.

2.3.9.6.2 Patient Blower

- Connect the MR patient blower supply to the ICC port labeled **PAT AIR SPY** on the lid (refer to Figure 2-17).
- Connect the MR patient blower supply to the ICC port labeled **PAT AIR RTN** on the lid (refer to Figure 2-17).
- Install air hose clamp connections until snug.

2.3.10 Magnet Monitor Installation*Figure 2-28: Magnet Monitor Installation Location on the ICC*

The Magnet Monitor (as described by GEHC 2394952PSP) shall be installed on either the left or right panels of the ICC. Six M4x0.7, 10mm long, pan head Phillips bolts are pre-installed onto either panel. Install the Magnet Monitor as follows:

1. Use a No. 2 Phillips drive to unfasten the Magnet Monitor Bolts (pre-installed) on the desired installation side.



2. Place the Magnet Monitor onto the panel and align with the mounting holes.
3. Reinstall the Magnet Monitor Bolts.
4. Route the Magnet Monitor flow meter cable through the opening on the lid labeled 14 as shown in Figure 2-17.

2.3.11 Cryogen Compressor Flow Meter Connection

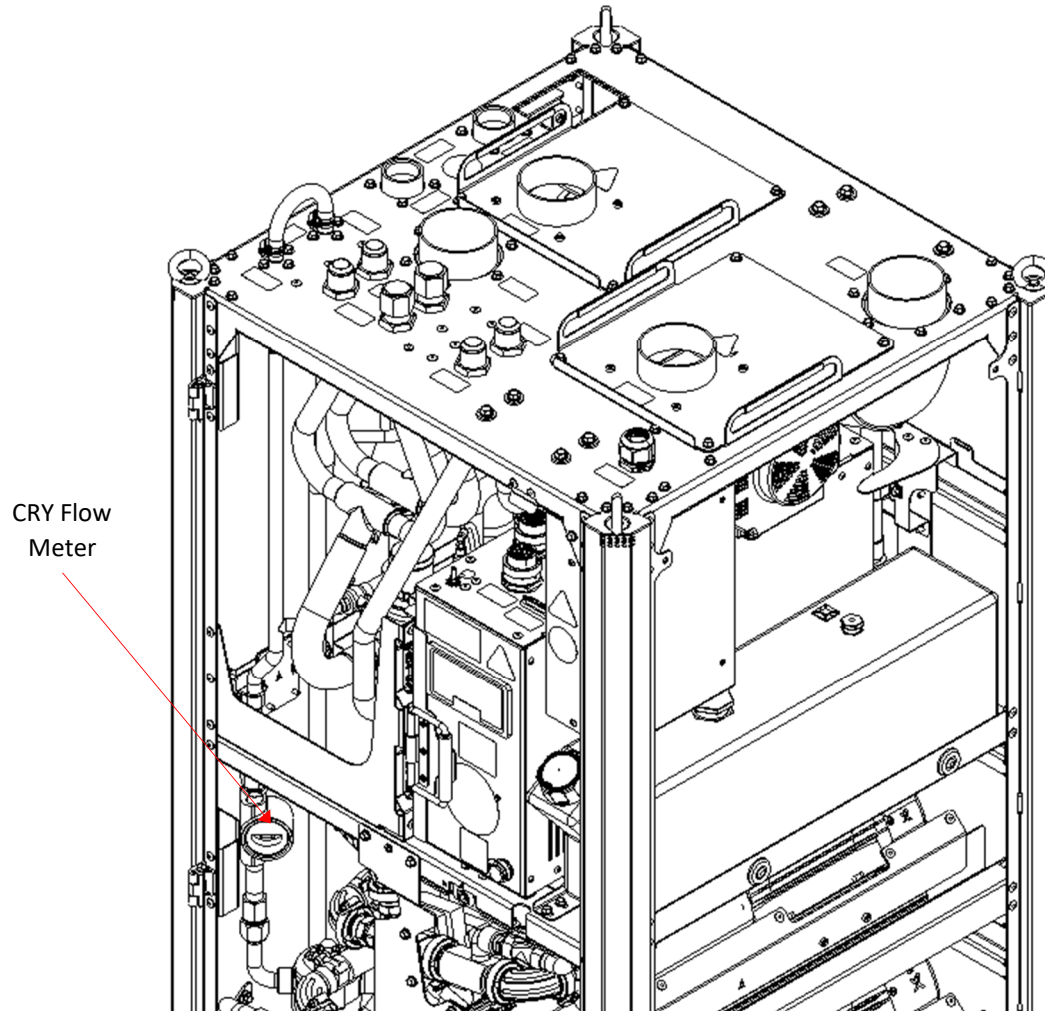


Figure 2-29: Cryogen Compressor Flow Meter

The cryogen compressor (CRY circuit) flow meter is electrically installed to the Magnet Monitor as follows:

1. Ensure that the Magnet Monitor has been installed according to GE 5804162 Magmon4 Operating and Service Manual.
2. Access the ICC via the front door (refer to 2.3.5).



Installation Guidelines

3. Locate the cryogen compressor flow meter (refer to Figure 2-29). Connect the previously routed Magnet Monitor flow meter cable to the 4-pin cable on the cryogen compressor flow meter.

2.3.12 Startup Process/Initial Priming/Re-priming

When the ICC is started for the first time (power and pump enable signals are present) the system automatically performs a priming sequence in the secondary circuits to purge air and fill the plumbing lines with coolant. To perform this process:

1. Ensure that incoming power connections have been installed according to 2.3.9.2 then turn on main incoming power from the Power Distribution Unit. This will automatically power the ICC and turn on the controller (refer to Figure 2-30). A lit screen indicates that the controller is powered, and power is present to the ICC.

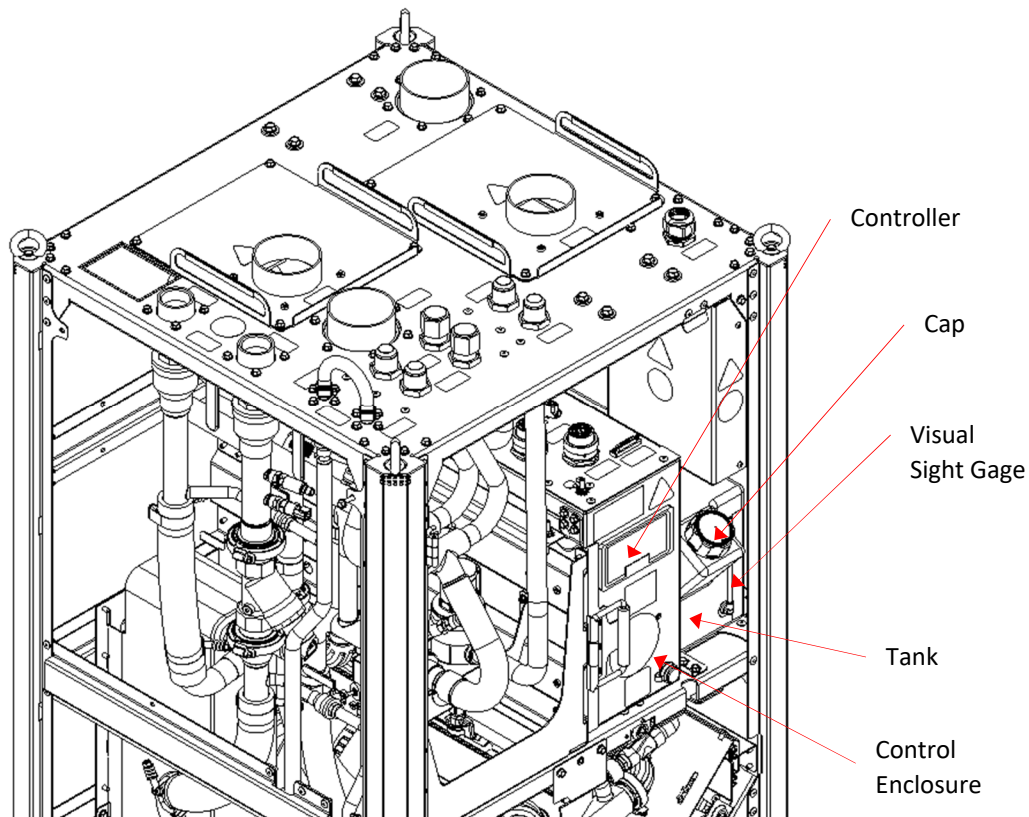


Figure 2-30: ICC Controller and Tank Location



NOTICE

If the ICC does not automatically start once power has been supplied ensure that the incoming power supply circuit breaker in the power distribution enclosure (refer



Installation Guidelines

to Figure 2-19) and all circuit breakers in the Control Enclosure (refer to Figure 2-30) are closed/turned on.

2. Wait for the controller to boot. The default homepage (refer to Figure 2-31) will be shown once the software has loaded.

	Out Temp	Setpoint
CCU	30.5%	30.0%
GCU	17.5%	17.0%
FPU In Temp:		10.0%

Figure 2-31: Controller Display Upon Startup



NOTICE

The FPU flow must be shut off during the startup/prime process to prevent condensation in the secondary circuits. The controller will not proceed with the startup/prime process if FPU flow is detected.

3. The correct operational mode will be configured by GEHC FE at the installation site.



NOTICE

The operational mode of the ICC can be changed only when both GCU and CCU pumps are stopped. ***The pump enable signals from the Cabinet Monitor must be disabled, the 44-pin D-sub Jumper must be disconnected, or both GCU and CCU pumps are disabled from HMI menu.***

4. Enable the pump start command to the ICC by either:
 - a. Connect the provided 44-pin D-sub Jumper (DTS PN 710205) connector to the J2 connector on the Control Enclosure (refer to Figure 2-32). ***The 44-pin D-sub Jumper is obtained from the Ship Loose Box (refer to 2.4).***
 - b. Connect the 44-pin D-sub cable that is connected to Cabinet Monitor (GEHC cable run number E3040). Refer to 2.3.9.3.



Installation Guidelines

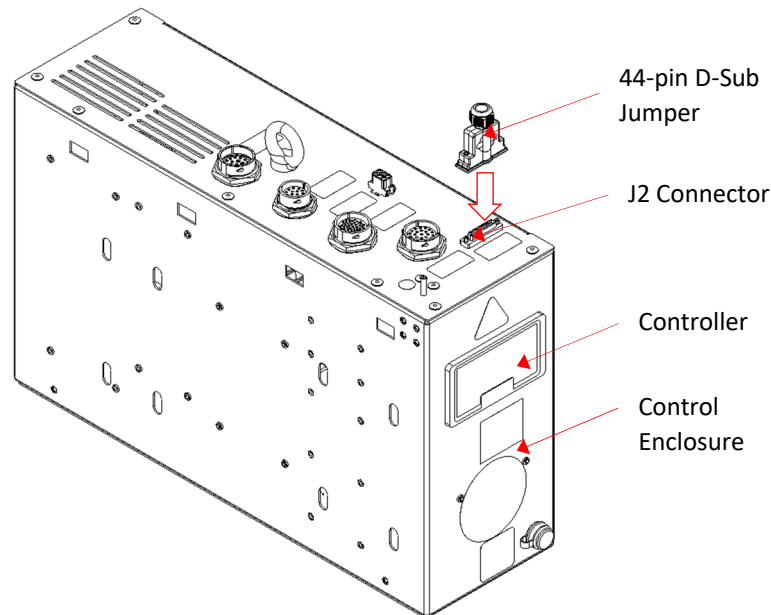


Figure 2-32: 44-Pin D-Sub Jumper Installation

5. The startup procedure will begin automatically. Follow the onscreen instructions to proceed. ***If FPU flow is detected the controller will prevent priming from occurring. Clear the alarm by holding on the  button on the controller for 3 seconds (refer to 3.1 to locate this button).***
6. Locate the tank Cap and uninstall it (refer to Figure 2-30).



NOTICE

The tank coolant must be filled with **GEHC PN 5174313**. All other alternative coolants must be approved by GEHC prior to use.

7. Fill the tank up with the provided coolant up to its **Filling Zone** shown on the **Tank Level Indicator Label**. The tank level can be monitored by the visual level gage (refer to Figure 2-33). ***A filling Funnel with Screen is provided in the Ship Loose Box for easier filling of the tank (refer to 2.4).***



Installation Guidelines

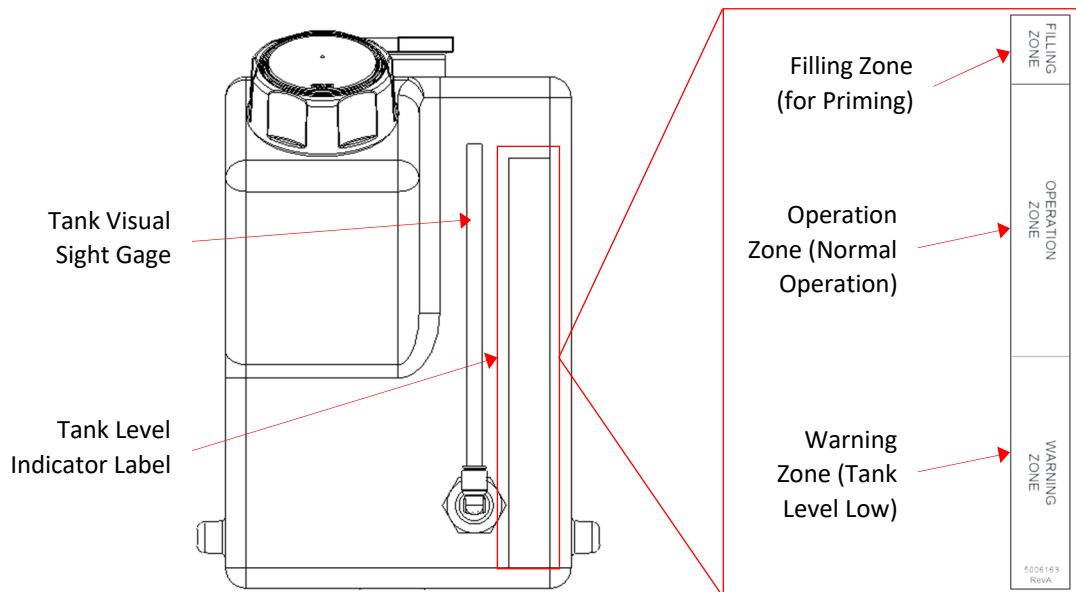


Figure 2-33: Tank Front View



CAUTION

Do not overfill the tank. Excess coolant will be discharged via the visual sight gage onto other internal components or the floor.

8. The controller will automatically begin the priming procedure of each pump.
 - a. As system undergoes this process, the coolant volume in the tank will gradually decrease. The controller will pause the priming process and prompt for additional coolant if there is insufficient coolant in the tank. The process will resume automatically when there is sufficient coolant. Repeat step 7 as prompted by the controller until the priming process is complete. Ensure that the coolant level is kept in the **Filling Zone** throughout the priming process.
 - b. If a fault is encountered, an error message will be displayed. Correct this fault then hold the  button on the controller for 3 seconds to reset the alarm and attempt to continue the priming process (refer to 3.1 to locate this button on the controller).
 - c. While the system is undergoing the priming process visually inspect newly installed or modified FPU, CCU and GCU connections to ensure that they are free of leaks. Fix leaking connections before proceeding.
9. Reinstall the tank Cap until hand tight.
10. Once the Startup Procedure has been completed, the controller will prompt for FPU flow to be present. Enable flow:

- a. If the FPU valves were closed, fully open the FPU Supply Valve (refer to Figure 2-27, handles parallel to flow).
 - b. If flow was stopped/diverted by the FPU cooling supply source, re-enable flow to the FPU circuit.
11. Use either the  or  button to select **Yes** (refer to 3.1 to locate these buttons on the controller).
 - a. Pressing the  button when **No** is selected will restart the priming procedure the next time pump-enable signals are present.
12. Disable the pump start command to the ICC:
 - a. If the 44-pin D-sub Jumper connector was used, remove it, and proceed to install the 44-pin D-sub serial cable (refer to 2.3.9.3).
 - b. If the 44-pin D-sub serial cable was installed and used, disable this signal from the Cabinet Monitor (refer to GEHC instructions).
13. **Perform the FPU-CRY pressure balancing procedure as described in 2.3.13.**
14. Clean up any spilled coolant.
15. Close and lock the front door if no further actions are required.

**NOTICE**

If the Startup Procedure has been aborted any time before completion, by disabling the pump enable signals, the controller will automatically restart the Startup Procedure automatically the next time the pump enable signals are present.

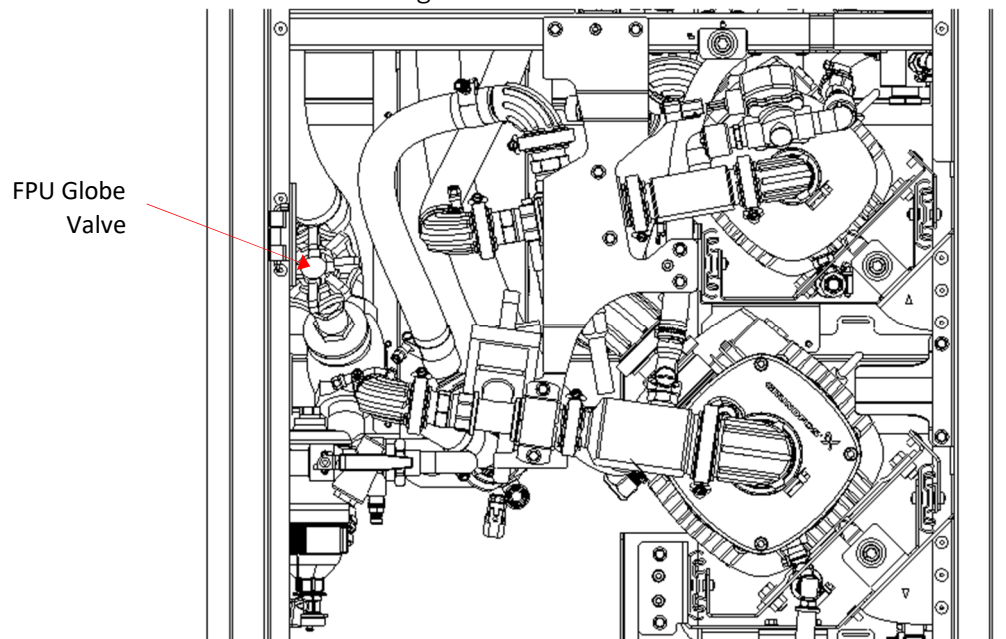
2.3.13 FPU-CRY Circuit Pressure Balancing

Figure 2-34: FPU Globe Valve



Installation Guidelines

2.3.13.1 Low Flow Application Installation

The ICC is configured for low flow applications from the factory. No further adjustment is required for the FPU Globe Valve.

2.3.13.2 Other Flow Mode Application Installation (High Flow, Artist-U, Artist-R)

1. Obtain the FPU Globe Valve handle from the ship loose kit (refer to 2.4).
2. Install the handle onto the valve (refer to Figure 2-34):
 - a. Uninstall retaining nut over valve stem.
 - b. Install the FPU Globe Valve handle.
3. Fully open the valve by turning counterclockwise.

2.3.13.3 Switching between Flow Modes for Existing Installations

If the existing installation is switching from **any other mode to Low Flow**:

1. Ensure that the ICC is supplied with incoming power.
2. Ensure that the cryogen compressor flow meter is connected to the Magnet Monitor (refer to 2.3.11).
3. Ensure that the Magnet Monitor is powered on.
4. Ensure that there is coolant flow on the primary side (entering and leaving the FPU circuit). This is done by monitoring FPU flow rate on the controller (refer to 3.3).
5. Ensure that the CRY supply and return valves are configured correctly in accordance with section 2.3.9.5.2.
6. Obtain the FPU Globe Valve handle from the ship loose kit (refer to 2.4).
7. Install the handle onto the valve (refer to Figure 2-34).
8. Fully open the FPU Globe Valve (refer to Figure 2-34) by turning the handle counterclockwise.
9. While monitoring the cryogen compressor coolant flow rate on the Magnet Monitor, gradually close the valve by turning the valve handle clockwise until the flow rate reads at least 8.5 LPM.



NOTICE

Whenever the facility plumbing on the primary side (to FPU) has been modified (length of plumbing changed, additional components added, etc.) all steps in this section must be repeated to ensure sufficient flow rate to the cryogen compressor.

If the existing installation is switching from **low to high flow**, refer to 2.3.13.2.

2.4 Ship Loose Kit

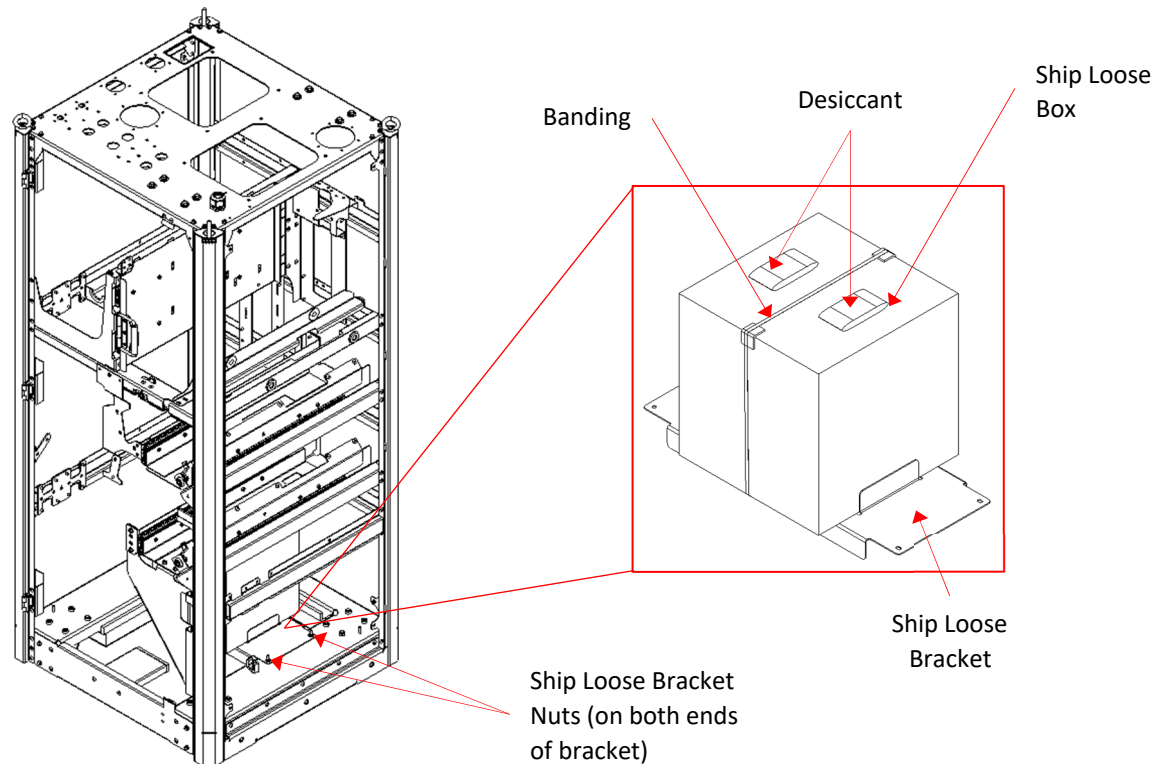


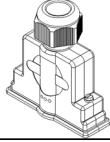
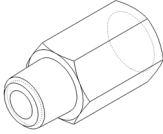
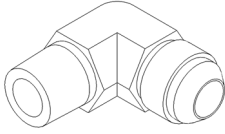
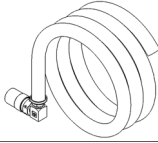
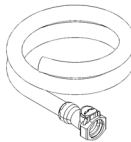
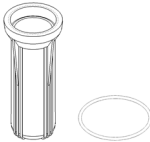
Figure 2-35: Ship Loose Box

The ICC is shipped with an accessories box that contains shipped-loose items (components generally installed at the final installation location or used for service). The Ship Loose Box is banded onto the Ship Loose Bracket that is mounted in the cryogen compressor cavity within the ICC. To obtain this box:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Use a 10mm hex wrench to unfasten the four Ship Loose Bracket Nuts to uninstall the Ship Loose Bracket.
3. Remove the Ship Loose Bracket from the ICC.
4. Carefully cut the banding that attaches the Ship Loose Box to the Ship Loose Bracket.
5. Remove the attached Desiccant bags on the top of the Ship Loose Box and discard. The Ship Loose Bracket and Nuts can be retained for future use.

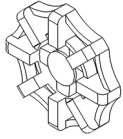
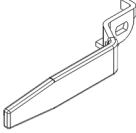
The Ship Loose Box contains the following components:

**Installation Guidelines***Table 4: List of Ship Loose Components*

Image	Description	Quantity
	J2 44-pin D-sub Jumper	1
	Cryogen Compressor Cooling Bushing	1
	Cryogen Compressor Cooling Elbow	2
	Filling Funnel with Screen	1
	Main Drain Hose	1
	CCU Tank Drain Hose	1
	GCU Tank Drain Hose	1
	Dirt Separator Drain Hose	1
	Filter Housings with O-rings	2



Installation Guidelines

Image	Description	Quantity
	FPU Balancing Globe Valve Handle	1
	CRY Flow Setter Valve Handle	1

2.5 Disposal

2.5.1 Coolant

All coolant to be disposed shall be done according to local regulations and guidelines.

2.5.2 Subcomponents and ICC

GEHC is the responsible party for take-back per WEEE requirements. All subcomponents of the ICC, and the ICC itself, shall be sent to GEHC for disposal according to GEHC's disposal guidelines. Contact GEHC for further information on this process.

3 Controller

3.1 Layout

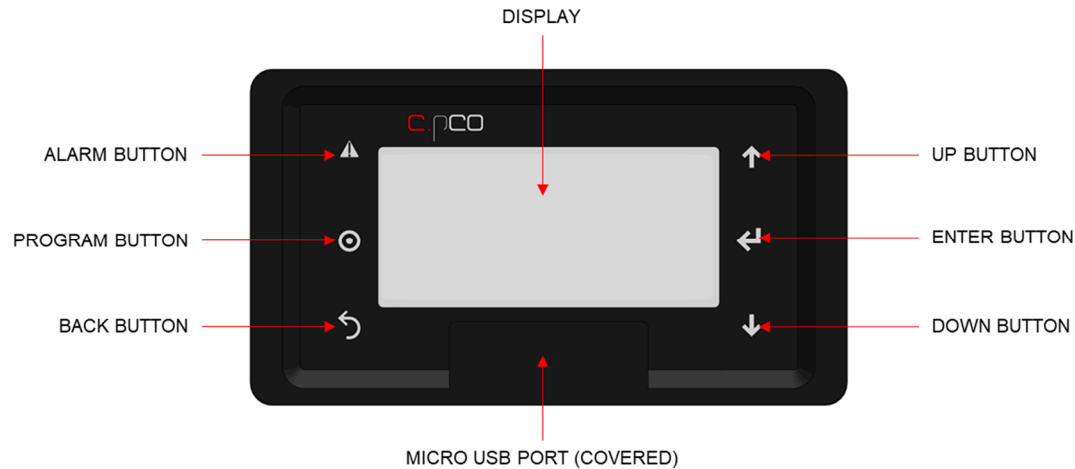


Figure 3-1: ICC Controller Layout

3.2 Navigation

To navigate through the controller's screens:

- At the main menu level
 - Use the  and  buttons to select the desired option.
 - Press the  button when the desired option has been selected to access it.
- Within a menu selection
 - Press the  or  buttons to navigate through each of the available screens.
 - To change the value of a setting (if permissible)
 1. Select the desired screen.
 2. Press the  button until the desired field has been highlighted (noted by a blinking cursor).
 3. Press the  or  buttons to increase, decrease or toggle the field's value.
Note: For numeric fields holding the  or  buttons will accelerate the rate of increase or decrease of the value, respectively.
 4. Press the  button to confirm the field's current value or press the  button to cancel changes.
 - Press the  button to return to the previous menu.
- Alarms
 - Press the  button to access the alarm menu.
 - If at least one alarm is active:
 - The  button on the controller will be lit red if a warning or fault is present.

- If more than one alarm is currently active press the  or  buttons to navigate through them.
- On the Alarm Log screen press the  button to access all previous alarms. Use the  and  arrows to navigate through them.
- Press the  button as required to return to the main menu.

3.3 Main Menu

The following is the layout of the main menu and submenus:

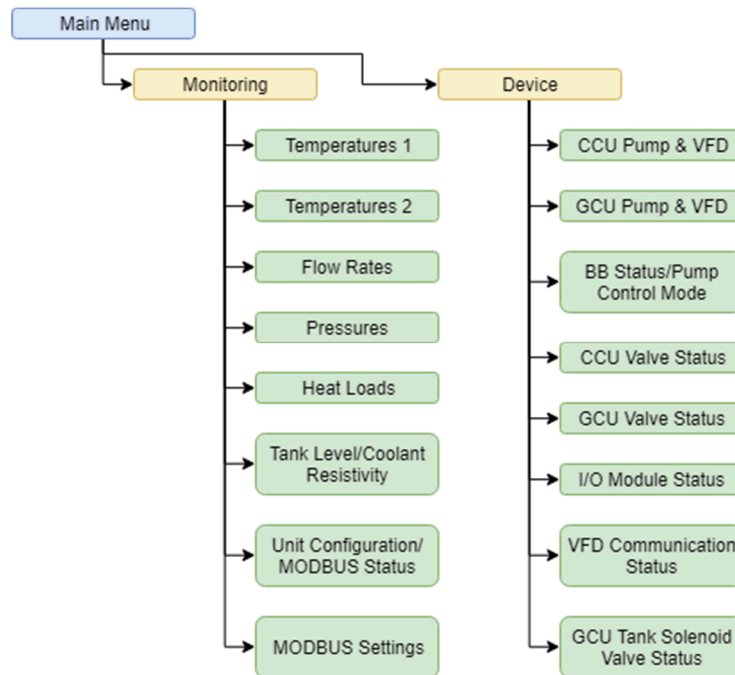


Figure 3-2: Main Menu Tree

4 Operation

4.1 Normal Operation

All normal operation is handled by signals from the Cabinet Monitor via the 44-pin D-sub serial cable. No intervention from the operator is required.



WARNING

Always ensure that the front door is properly closed and securely fastened when the ICC is operating under normal operating conditions to prevent unauthorized access into the ICC.

4.2 Off-Normal Operation

All off-normal operation must be approved by both GEHC and the factory.



4.3 Coolant Maintenance

4.4 Secondary Cryogen Compressor Cooling (City Water Bypass)

If the FPU cooling source has been compromised, (failure of chiller or cold-water supply, or during a service operation on the primary side) cooling must still be provided to the cryogen compressor (CRY circuit). The secondary source is typically a water supply provided at the facility (such as from city service) but can be any other source that meets the minimum cooling requirements of the cryogen compressor. Always refer to the cryogen compressor's manual for this information.



WARNING

Ensure that the facility supply has been shut off or is able to tolerate a no-flow condition in the ICC as the FPU supply and return valves will need to be closed to facilitate cooling from the secondary source.



WARNING

The supply pressure for the secondary cooling source, measured at the inlet of the ICC, must **NOT** exceed **6 bar**.



WARNING

The supply temperature for the secondary cooling source, measured at the inlet of the ICC, must be 5 to 15 °C for high flow mode or 5 to 12°C for low flow mode.

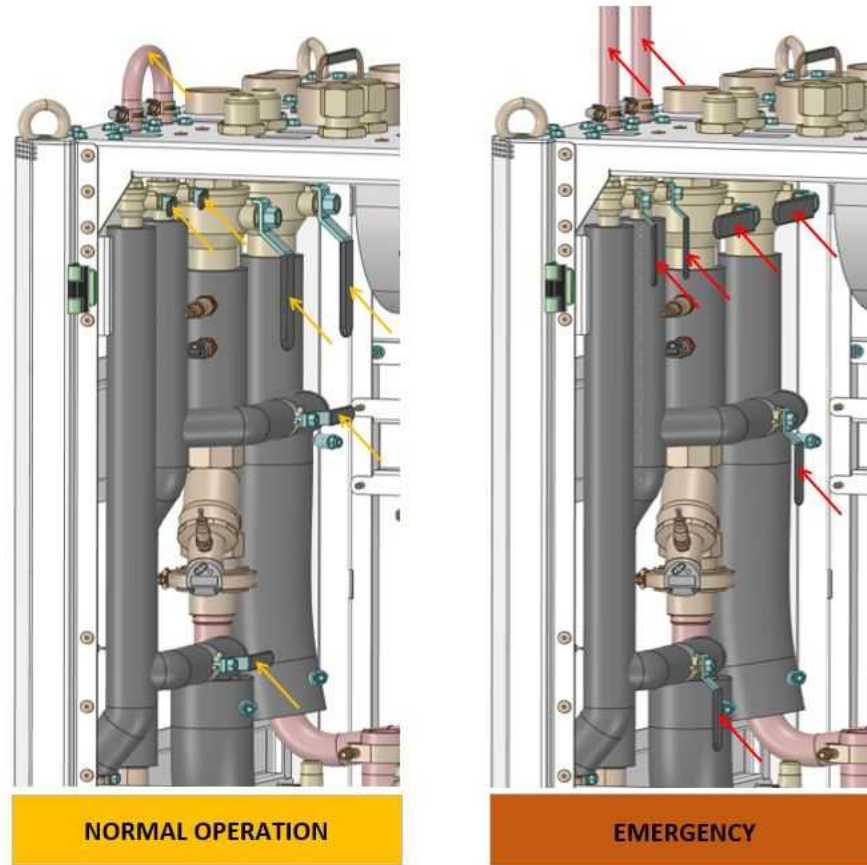


Figure 4-1: City Water Valve Configuration (on right)

To swap cooling over to city water (or an alternative cooling source):

1. Access the ICC through the front door (refer to 2.3.5).
2. Close both FPU supply and return valves (refer to Figure 2-27, handle perpendicular to direction of flow).
3. On the lid, locate the ports labeled CW SPY AND CW RTN (refer to Figure 2-17). If a jumper hose that connects both these ports is present:
 - a. Use a 5/16" wrench to unfasten the two hose clamps.
 - b. Remove the jumper hose.
4. Install the secondary CRY cooling connections as follows:
 - a. Ensure that the secondary source has been shut off upstream (or at the in ICC, refer to Figure 2-26).



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- b. Slip the previously removed, loosely fastened hose clamps over each (supply and return) city water hose. The hoses should be nominally 1/2" (12.7mm) ID and have an OD no greater than 31.8mm.
 - c. Connect the secondary cooling supply to the ICC port labeled **CW SPY** on the lid (refer to Figure 2-17). Connect the secondary cooling return to the ICC port labeled **CW RTN** on the lid (refer to Figure 2-17). Ensure that the edge of each hose is pushed past the hose bead on the tube by at least 25mm.
 - d. Tighten the two hose clamps.
5. Configure the CRY and CW valves as follows (valve configuration label also provided on the inside of the ICC door as shown in Figure 4-1):
 - a. Close both CRY supply and return valves (refer to Figure 2-26, handle perpendicular to direction of flow).
 - b. Open both CW supply and return valves (refer to Figure 2-26, handle parallel to direction of flow).
6. Turn on the secondary cooling source upstream. Ensure that the coolant flow is sufficient (8.5 ± 1.5 LPM) through the cryogen compressor by monitoring the Magnet Monitor. Adjust this flow rate at the source as necessary.
7. Close and lock the front door if no further actions are required.

To revert to primary cooling in the FPU:

1. Access the ICC through the front door (refer to section 2.3.5).
2. Turn off the secondary cooling source upstream.
3. Ensure the primary cooling source is turned off upstream.
4. Configure the CRY and CW valves as follows (valve configuration label also provided on the inside of the ICC door as shown in Figure 4-1):
 - a. Close both CW supply and return valves (refer to Figure 2-26, handle perpendicular to direction of flow).
 - b. Open both CRY supply and return valves (refer to Figure 2-26, handle parallel to direction of flow).
5. Open both FPU supply and return valves (refer to Figure 2-27, handle parallel to direction of flow).
6. Turn on the primary cooling source.
7. If the primary cooling circuit has changed perform the FPU-CRY circuit balancing according to 2.3.13.
8. Use a 5/16" wrench to uninstall the hose clamps of the secondary cooling hoses.
9. Slip the hose clamps past the hose beads on the CW SPY and CW RTN tubes and fasten until they are no longer loose to prevent misplacing them for future use.
10. Close and lock the front door if no further actions are required.



Operation

4.4.1 Fluid Level Check

The tank level can be monitored by the visual level gage (refer to Figure 4-2). check that the coolant level is within normal operation mode. Fill the tank up with the provided coolant, FRU part number 5174313-4, up to its Filling Zone shown on the Tank Level Indicator Label. Estimated time for this planned maintenance procedure is 10 minutes. A filling Funnel with Screen is provided in the Ship Loose Box for easier filling of the tank.

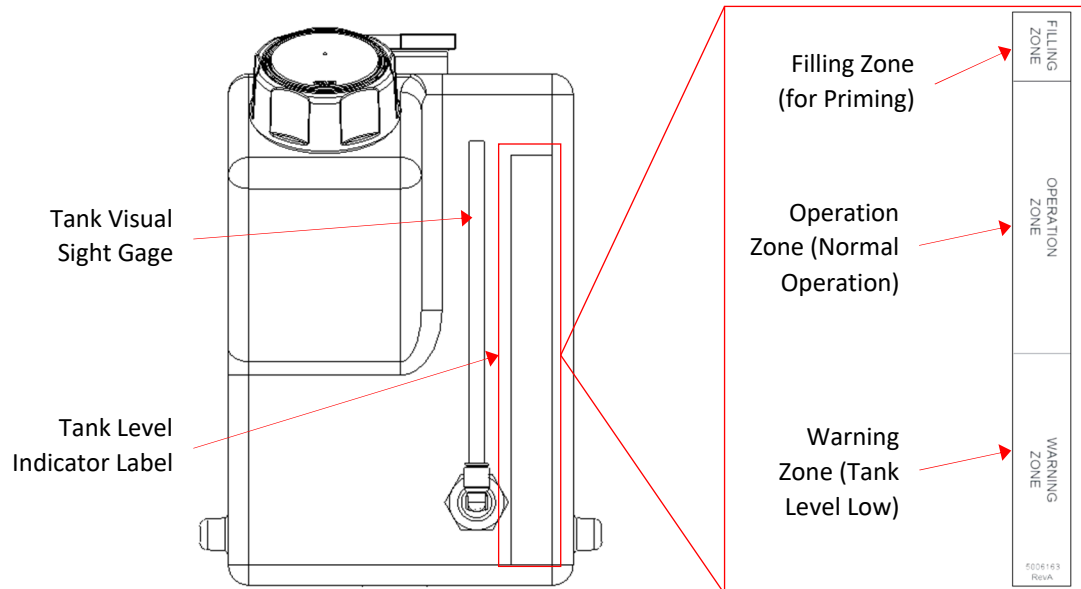


Figure 4-2: ICC Tank Visual Gage Indicator

4.5 Field Replaceable Units (FRUs)

A Field Replaceable Unit (FRU) is defined as a component or subassembly within the ICC that is intended to be replaceable in the field (at installation site) due to a failure.



4.5.1 List of FRUs

Table 5: List of ICC Field Replaceable Units (FRUs)

FRU	GEHC PN
CCU Pump	5806842-2
GCU Pump	5806845-2
CCU Flow Switch	5875310
GCU Flow Switch	5875311
FPU Flow Meter	5806849-2
CRY Flow Meter	5806848
CCU Mixing Valve	5806844-2
GCU Mixing Valve	5806847-2
CRY Flow Setter (Cartridge)	5817844
Control Panel Enclosure	5806852-2
Power Distribution Enclosure	5818155
Body Coil Blower	5806850
Patient Blower	5806851
CCU Supply Temperature Sensor	5820743
CCU Return Temperature Sensor	5820743
GCU Supply Temperature Sensor	5820743
GCU Return Temperature Sensor	5820743
FPU Supply Temperature Sensor	5820743
CCU Supply Pressure Transducer	5820744
FPU Supply Pressure Transducer	5820744
GCU Supply Pressure Transducer	5820768
Brazed Plate Heat Exchanger (BPHX)	5817847-2
Tank	5806853
Dual DI Cartridge Bracket	5820741
CCU Pump Suction Manifold	5830105
GCU Pump Suction Manifold	5868485-02
Deionization Filter Kit	5264701
Gasket kit (1.5 in. and 1 in.)	5868484



Glen Dimplex Contact Information**5 Glen Dimplex Contact Information**

2625 Emerald Drive
Kalamazoo, MI 49001 USA
<https://www.dimplexthermal.com>

6 GE Healthcare Contact Information

3200 N. Grandview Boulevard
Waukesha, WI 53188 USA
<https://www.gehealthcare.com>



7 Glossary

Table 6: List of Acronyms and Definitions

BB	Body Coil Blower
BPHX	Brazed Plate Heat Exchanger
BSPT	(ISO) British Standard Pipe Tapered
CCU	Cabinet Cooling Unit
CRY	Cryogen Compressor
DI	Deionized
FLA	Full Load Amperes
FPT	Female (National) Pipe Thread
FPU	Facility Plumbing Unit
FRU	Field Replaceable Unit
GCU	Gradient Coil Unit
GEHC	General Electric Healthcare Company
GPM	(US) Gallons per Minute
GHT	(US) Garden Hose Thread
ICC	Integrated Cooling Cabinet
ID	Inner Diameter
ISC	Integrated Systems Cabinet
JIC	Joint Industry Council
LPM	Liters per Minute
MCA	Minimum Current Ampacity
MOPD	Maximum Over-current Protection Device
MR	Magnetic Resonance
NPT	(US) National Pipe Thread
NTC	Negative Temperature Coefficient
OD	Outer Diameter
OS	Operating System
PB	Patient Blower
PDU	Power Distribution Unit
PHPS2	Patient Handling Power Supply 2
PN	Part Number
PPE	Personal Protective Equipment
PTFE	Polytetrafluoroethylene
RTD	Resistance Temperature Detector
USB	Universal Serial Bus
VFD	Variable Frequency Drive
WEEE	Waste Electrical & Electronic Equipment



8 Appendix

8.1 NPT Pipe Installation Guidelines (ASME B1.20.1)

This section provides a guideline to installing NPT pipe connections per ASME B1.20.1 standards. Per section 1.4.3 of this referenced standard:

“Due to application-specific variables such as materials, wall thickness, operating pressures, etc., no guidance is given in this Standard regarding joint-tightening torque. However, joints should be tightened beyond the hand-tight engagement position. Advancing the joint past hand-tight creates interference between external and internal thread flanks, produces a seal (with the use of a sealant), and helps prevent loosening of the joint. Overtightening may be detrimental to the sealing function of the joint.”

The following table summarizes several common NPT pipe sizes and their recommended installation procedures:

Table 7: NPT Engagement Depth Recommendations

NPT Pipe Size ²	Hand Tight Engagement (Number of turns) ³	Wrench Make-up Engagement (Number of turns)
1/4	4	3
3/8	4.25	3
1/2	4.5	3
3/4	4.75	3
1	4.5	3
1-1/4	5	3
1-1/2	4.75	3
2	5	3

² Applicable to equivalent BSPT pipe thread sizes as well.

³ Rounded to the nearest quarter turn.

8.2 Manifold Flange Fitting Installation

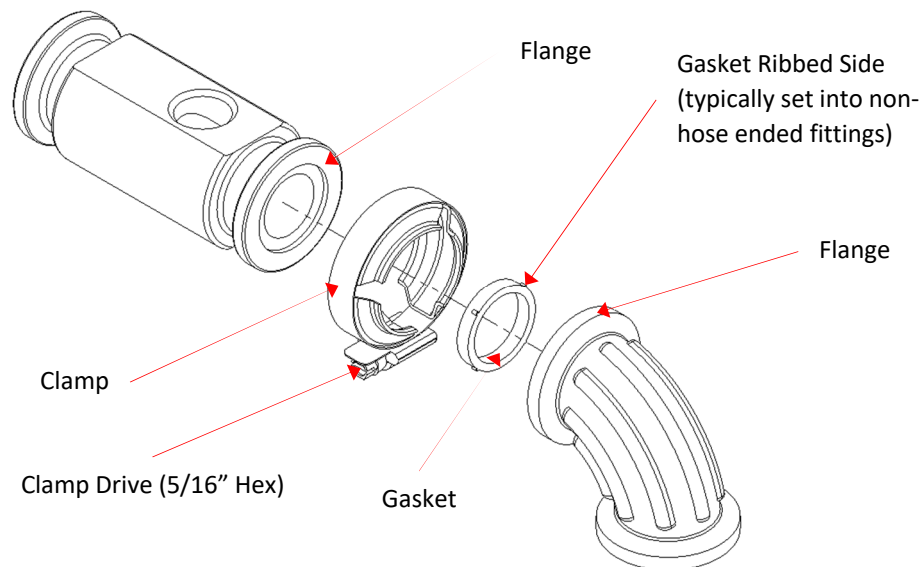


Figure 8-1: Flange Manifold Installation

To uninstall a manifold flange fitting:

1. Locate the flange clamp that connects the fittings that need to be separated.
2. Unfasten the flange clamp's tightening bolt using a female 5/16" (~8mm) hex drive until the clamp is sufficient loose to separate the two fittings. Slip the loosened clamp onto the flange of fitting that is not to be removed.
3. Separate the two fittings and retrieve the gasket in between.
4. **Discard used gaskets.** Replacement gaskets are supplied with each FRU kit or as part of the Manifold Flange Gasket FRU kit (GEHC PN 5868484).

To reinstall a Banjo manifold flange fitting:

1. Select the appropriate clamp and gasket for the flange fitting being reinstalled.
 - a. Use a 1" clamp and gasket for 100 Series sizes (smaller)
 - b. Use a 1-1/2" clamp and gasket for 200 Series sizes (larger)
2. Place a new gasket on into the gasket groove of the fixed (non-hose ended) fitting. Ensure that the side of the gasket with the ribs is placed into the gasket groove of the fixed fitting to hold it in place.
3. If the clamp has been completely unfastened, loosely fasten it by engaging the band under the worm drive and fasten the tightening bolt using a female 5/16" (~8mm) hex drive. Slip the loosely fastened clamp over the fixed fitting's flange.



Appendix

4. Align the mating flange fitting's gasket groove over the gasket that was previously installed in step 2 and firmly press both fittings together, ensuring that the gasket remains concentric to both fittings.
5. While pressing both fittings together reorient the loosely fastened clamp in step 3 over the flange feature of both fittings and fasten the tightening bolt of the clamp using a female 5/16" (~8mm) hex drive. Tighten the clamp to required torque and continue to observe fitting for leaks during priming or operation. If leaks occur, retighten as necessary until leaks stop.
 - a. Torque range for 1 in. clamp and gasket for 100 series (smaller) = 5.6 to 6.8 N-m.
 - b. Torque range for 1 ½ in. clamp and gasket for 200 series (larger) = 10.2 to 11.3 N-m.

Note: If leaks persist and no further tightening is possible, the gasket may be damaged or seated incorrectly. Uninstall the clamp, obtain gasket, and inspect. If the gasket is damaged, replace it.

8.3 Quick Connect Fittings

8.3.1 Parker Moldmate

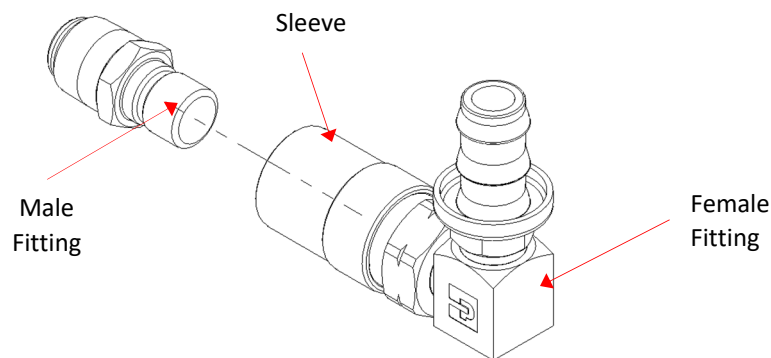


Figure 8-2: Parker Moldmate Connections

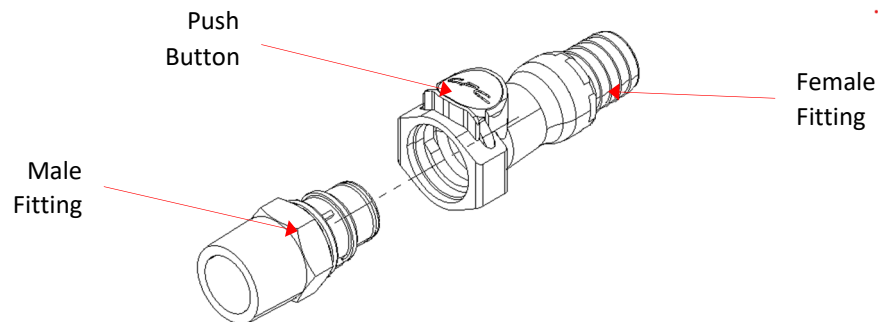
The Auxiliary Tank Return and Drain Hoses use Parker Moldmate fittings at one end.

To connect Moldmate fittings:

1. Retract the Sleeve on the Female Fitting and hold.
2. Connect the Female Fitting to the Male Fitting.
3. Release the Sleeve. Both fittings should hold in place.

To disconnect Moldmate fittings:

1. Retract the sleeve on the female fitting and pull away from the male fitting.

**Appendix****8.3.2 Colder Product Company (CPC)***Figure 8-3: CPC Connections*

The Tank-Pump Connections and Tank Drain Hoses use CPC fittings at one end.

To connect CPC fittings:

1. Push the male or female fitting over its mating fitting until push button clicks.

To disconnect CPC fittings:

1. Hold down on the Push Button on the Female Fitting.
2. Pull the two fittings apart then release the Push Button.



Revision History

9 Revision History

Table 8: Revision History

Revision	Date	Description of Changes	Author
1	19-Jan-2022	Initial Release	Frank Rinderspacher
2	17-Mar-2022	Updated external packaging	Frank Rinderspacher



Revision History

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